NIHR | National Institute for Health and Care Research

Summary

Reference Number	NIHR305253
Lead Applicant	Ms Helena Jackson
Research Title	Improving the detection and treatment of malnutrition for people with chronic kidney disease: co-design and validation of a novel outpatient nutrition screening tool.
Plain English Summary	**Title:** Improving nutritional care for people with chronic kidney disease: a new way to identify and prevent malnutrition. **Background** Protein energy malnutrition develops when food intake does not meet the body's needs, or the body does not respond in the normal way. Malnutrition affects up to half of people with more advanced chronic kidney disease, including those on dialysis. This has many causes including poor appetite, fatigue, treatment side-effects and dietary restrictions. Malnutrition can be distressing, make people feel weak and tired, develop medical problems and serious illness, and spend more time in hospital. As with cancer, looking for malnutrition earlier makes it easier to prevent and treat. It also reduces healthcare costs. Nutrition screening is the recommended way to look for malnutrition, guide care and decide whether to refer to a dietitian. To do this, nurses need the right 'tool', a standard set of questions leading to a care plan. General screening tools do not work well. Earlier attempts at kidney-friendly ones were too complicated and not tested in real-life settings. Patients and nurses were not asked for their opinions or to help develop them. We previously developed and tested a screening tool for kidney wards. We will work with nurses and people with kidney disease to adapt it for outpatient services. This will include developing training resources and self-management information and testing these in three hospitals. **Aim** To work with nurses and people living with kidney disease to review and adapt an existing inpatient nutrition screening tool for outpatient services. **Methods** We will ask nurses and people living with kidney disease to take part. They will be from three large kidney units that serve communities with a range of social and ethnic backgrounds. **Study 1.** Research review To review what is known about nurse-led nutrition screening in kidney outpatient services, the tools used and views of nurses and service users. This will identify discussion topics and priorities for the codesign study. **Study 2. Codesign** To adapt the inpatient screening tool for outpatient settings with nurses and people with kidney disease. This will include planning how it is used and designing resources for training and self-management through several rounds of individual and group meetings. **Study 3. How it works in practice** We will introduce the new screening tool into specialist kidney outpatient services for people with advanced kidney disease and on dialysis. We will check that the new screening tool works well in practice, using a recommended method. Nurses will complete it with patients as usual and give them some self-help information. Some patients will be asked the same questions by another staff member to check this gives the same result. For 139 patients who agree to take part, a kidney dietitian will do a detailed assessment and test handgrip strength to check for malnutrition and compare with the screening tool. Participants will be asked to use and rate the resources. We will ask participants and nurses for their opinions using a questionnaire. We will also assess whether having the screening tool helps nurses to include nutrition in care plans, give information and refer to dietitians. **Conclusion** This project fits with NHS goals for better patient care and quality of life and less need for hospital treatment and admission. It will provide the first practical, tested specialist nutrition screening tool for kidney outpatient services, designed with nurses and

	people with kidney disease. This will help to improve access to nutritional care for more people who need it. This includes future use of hospital electronic systems and online platforms. It will also help to improve future healthcare and malnutrition research.

Our purpose for collecting this data is to communicate with you about your application and have the necessary information to evaluate you for a grant. The data we collect here is collected in the public interest. Information provided here may be subject to Freedom of Information requests.

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1. Application Summary Information

Host organisation (which will administer any award)
St George's University Hospitals NHS Foundation Trust

Partner organisation	Kingston and St George's Faculty of Health, Social Care and Education

Research Title	Improving the detection and treatment of malnutrition for people with chronic kidney disease: co-design and validation of a novel outpatient nutrition screening tool.

Partnership Fellowship Please select the relevant Partnership Fellowship.
N/A

Proposed WTE Please select which WTE option you would prefer
0.80

Duration (months) The duration for your award is based on your chosen WTE
45

Proposed start date if grant awarded	01/04/2025

Research Type	Pilot study

2. Applicant CV

A CV for the Applicant is a mandatory component of this application; please ensure this is completed fully.
Please note that some of the CV questions are automatically completed from information you have provided in the 'Update CV' section of the 'Manage My Details' page. These can be updated via the ARAMIS Home Page or by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

Degrees and Professional Qualifications

From	To	Qualification	Subject	Country	Institution	Class
09/1997	07/1999	MSc	Clinical Nutrition	United Kingdom (England)	University of Surrey	Distinction
09/1991	12/1992	Postgraduate Diploma	Dietetics	United Kingdom	University of London	Pass
09/1988	07/1991	BSc(Hons)	Nutrition	United Kingdom	University of London	1st

Present and previous positions
(including from/to dates, job title and employer)

From	To	Position	Organisation	WTE % Research
10/2003	-	Highly Specialist Renal Dietitian (Band 7)	St George's University NHS Foundation Trust	0
10/2002	10/2003	Lead Dietitian for Renal and Liver Services (Band 8a)	Kings College Hospital NHS Trust	0
09/1995	09/2002	Highly Specialist Renal Dietitian (Band 7)	Kings College Hospital NHS Trust	0
07/1993	09/1995	Liver Services Dietitian (Band 6)	King's College Hospital NHS Trust	0
01/1993	06/1993	Dietitian General Services (Band 5)	King's College Hospital NHS Trust	0

Research Grants held

Title of Grant	Role in the Research Grant	Total Grant £	Source of Grant (Organisation Name)	Duration (In Months)	Start Date of Funding	End Date of Funding
British Renal Society/ British Kidney Patients Association Research Grant Award	As lead applicant, I conceived the study, created the research team and collaborations. In the funded time I trained nurses and completed the study activities at 3 hospitals Outside the funded time I completed ethical applications, analysis of results conference presentations and research publication.	18488	British Renal Society/ British Kidney Patients Ass	6	01/11/2014	01/05/2015

Publication Record

This should display the authors, date, article title and journal name for each publication added. This information can be edited via the “My Research Outputs” page of your ARAMIS account.

Publications
Jackson, H.. (2023) ‘Enhancing nutrition screening in patients with kidney disease.’, *Nursing standard (Royal College of Nursing (Great Britain) : 1987)*, 38(9).
Brown, TJ., Williams, H., Mafrici, B., Jackson, HS., Johansson, L., Willingham, F., McIntosh, A. and MacLaughlin, HL.. (2021) ‘Dietary interventions with dietitian involvement in adults with chronic kidney disease: A systematic review.’, *Journal of human nutrition and dietetics : the official journal of the British Dietetic Association*, 34(4).
Jackson, HS., MacLaughlin, HL., Vidal-Diez, A. and Banerjee, D.. (2018) ‘A new renal inpatient nutrition screening tool (Renal iNUT): a multicenter validation study.’, *Clinical nutrition (Edinburgh, Scotland)*, 38(5).
Naylor, HL., Jackson, H., Walker, GH., Macafee, S., Magee, K., Hooper, L., Stewart, L. and MacLaughlin, HL.. (2013) ‘British Dietetic Association evidence-based guidelines for the protein requirements of adults undergoing maintenance haemodialysis or peritoneal dialysis.’, *Journal of human nutrition and dietetics : the official journal of the British Dietetic Association*, 26(4).

Relevant Prizes, Awards and other Academic Distinctions

Please note the information for additional prizes or awards is automatically populated from information provided in the "Update CV" section of the "Manage My Details" page, which you can update by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

Date (mm/yyyy)	**Source**	**Prize**	**Role of Applicant**
01/05/2023	Translational and Clinical Research Institute (St George's Hospital)	Senior Research Fellowship (competitive award): equivalent to 0.1WTE for 1 year	Holder of fellowship award
01/10/2018	St George's University Hospitals	AHP Innovator Award for developing the iNUT nutrition screening tool.	Renal Dietitian
01/06/2015	The British Renal Society	British Renal Society Recognition Award (Best Poster)	Lead author
01/10/2014	British Renal Society/British Kidney Patients Association	Research Grant	Primary Investigator

ORCID iD 0000-0003-3804-7317

3. Applicant Research Background

Professional Background	Allied Health Professional

Profession (Allied Health Professional)	Dietitian

Do you have current / active professional registration? This registration must be with one of the [ICA approved regulatory bodies]
Yes

Professional registration number/pin Please supply a valid professional registration number/pin followed by the regulatory body (eg 12A3456B - NMC or PA12345 - HCPC)
DT05662 - HCPC

Please describe your research career to date

Research and academic experience

At King's College Hospital (KCH), with a research-active dietetic department, I became interested in nutritional assessment in liver and kidney disease. I was involved with nutritional supplement trials and the early introduction and evaluation of intra-dialytic exercise. For my part-time MSc in Clinical Nutrition, I conceived and completed research on dialysis protein losses, achieving the only distinction, whilst working full-time. I participated, as the only non-medic, in an invitation-only international conference for young researchers. I presented at an invitation-only meeting at the Karolinska Institute, Stockholm, a leading centre for renal nutrition research.

At St George's Hospital, I initiated research collaborations with Surrey University on outpatient nutrition assessment and screening. Other university collaborations include patient information and self-management. I have led the development, piloting and evaluation of a renal-specific inpatient screening tool (iNUT) based on available evidence and in collaboration with nursing staff. I devised a multicentre research study, created research collaborations across three hospitals and was awarded a British Renal Society research grant (£18,000) . The ethical application, proposal and eventual publication (Clinical Nutrition: IF 6.4) were completed in my own unfunded time.

Subsequently, I have supported the introduction of iNUT into the heart failure and liver units, mentoring specialist dietitians with conference presentation and journal publications. I have developed guideline appraisal, systematic searching and critical appraisal skills, as well as evidence synthesis skills through involvement in two systematic reviews guiding current renal dietetics practice.

I have extensive experience in presenting at local, national and international conferences. Work with the British Dietetic Association Renal Nutrition Guidelines Group includes peer-reviewed evidence-based guidelines on dialysis protein requirements and a systematic review of dietetic interventions.

I have developed excellent writing skills for a lay and professional audience, including journals and websites, textbooks and as lead author for two cookery books. Recently I wrote on malnutrition in heart failure/kidney disease for nutrition health professionals (Complete Nutrition) and an invited peer-reviewed article on renal nutrition screening (Nursing Standard).

Through collaboration with academic institutions I have successfully co-supervised medical and dietetic students, including five MSc students to completion, gaining experience in supervision, study design, and providing feedback. I have developed research management skills through leading a multicentre study and as site lead on a PhD study. I am continuing to develop my research skills and network by participating in national projects on a dietetic complexity scale and on dietetic outcomes.

Research leadership

Dietetic profession

As a Band 7/8a at KCH I initiated a role for dietetics in the annual renal unit audit cycle to include more reliable nutritional assessment data. This included collecting and analysing the data and presenting at the annual audit meeting and at national meetings and setting standards for the outpatient dietetic service. This has continued at KCH to the present time.

At SGH I have led the renal dietitians in collaborative research and audit and local renal unit and dietetic standards and protocols, including intradialytic nutrition, renal bone disease and nutritional support. I lead the SGH Dietetics Department Research & Audit Workstream. We are currently working using quality improvement processes to upskill the department members and increase confidence and research output. Objectives include training, representation at the Trust Clinical Audit Day and utilising social media.

I regularly advise nutritional researchers in the UK and multiple countries on renal nutrition screening and implementation of the iNUT screening tool.

Renal Nutrition Group (RNG) of the British Dietetic Association

I have been an active volunteer of the Nutrition Guideline group and promoted these at dietetic and multi-professional wider profession, at regional, national and international meetings. Recently I have provided peer-review and written on behalf of the RNG for a patient charity. I represent the RNG on a national renal guideline group.

European Dialysis and Nurses Association/European Renal Care Association (EDTNA/ERCA)

Voluntary positions included Nutrition Group Co-chair and UK conference committee member. Experience included abstract peer review, promotion, presenting and chairing at international conferences in Europe and North America. I led the writing and publication of European Nutrition Guidelines including analysing surveys of European dietetic practice revising existing UK guidelines, and promoting the guidelines.

UK Kidney Association (UKKA) Living Well with Kidney Disease Special Interest Group Co-Chair.

This voluntary position involves organising and chairing meetings, working with clinicians, clinical researchers and academics to promote and evaluate aspects of clinical care that relate to the well-being of people with CKD. I have joined the Kidney Transplant Recipient guideline review committee. I completed a Royal Society of Medicine Leadership course in 2023 which developed my management skills and confidence in taking on new roles and challenges including recent research fellowship applications.

Education and teaching development

I have developed teaching skills through university lecturing on renal nutrition and nutrition assessment (including Surrey, London Metropolitan and King's College London Universities).Teaching on post registration renal nursing and dietetics specialist training courses with (Kingston and Surrey Universities) helps to support my evidenced-based practice through literature review and peer discussion and to refine my presentation skills, including virtual and interactive approaches. I regularly mentor staff and students in research, audit and writing for publication and conference presentation.

Career trajectory

I have become widely respected within my speciality but always strive to increase my knowledge, skills and patient care. I feel that I have now reached the limits of what I can achieve in research and my clinical career without further formal development and qualification that the NIHR DCAF offers. I ambitious to become a clinical academic within the planned St Helier/St George's regional renal research unit and to also lead nutrition screening, nationally and internationally. This includes supporting more renal and other specialist units to introduce and evaluate our screening tools, collaboration on research into nutrition screening and high-quality intervention studies of outcomes and nutrition support - all which are currently lacking, particularly for people with kidney disease and those with multiple medical conditions. I am keen to continue to support dietitians and other non-medical professionals in research and develop the expertise to do this more effectively.

Have you already registered for a PhD?	No

Has this application, or a similar application, previously been submitted to this or any other funding body?
Yes

If Yes, provide details
Submitted 2023 to Kidney Research UK (KRUK) and reached interview stage. The process included passing an expression of interest application, proceeding to a full application, a 2000 word rebuttal to a review by a panel of professional and lay experts and online interview. I have utilised the process and panel feedback for the current proposal, including substantial improvements to the codesign study, restructuring and expanding the research plan and trying to improve the communication of the importance of the work to non-nursing/dietetic professionals and lay members. I have also adjusted my supervisory team by the inclusion of Prof Fiona Jones as primary supervisor. I feel this has significantly strengthened the proposal and opportunities for my learning and development, particularly in self-management research, participatory methods, codesign and implementation science. She also provides a valuable perspective outside the specialisms of renal nutrition and nutrition screening and provides an excellent role model for my career aspirations. I have had further advice from the Research Support Service (RSS) on writing the detailed research plan including the structure, language and statistics. I have had further advice from a statistician, Dr Sarah White who will will be supporting me throughout the project, particularly in the validation stage and collaborating on research publications. I have also strengthened my PPI by meeting with KRUK and the RSS and having further lay review of my proposal through their expert advisors. I have greatly appreciated all the feedback and I have better understanding of the support that is available and will work KRUK PPI advisors throughout the fellowship.

Contextual Factors Please use this question to detail any contextual factors you wish to make the Selection Committee aware of. NIHR wants to know about any circumstances so that they may take them into consideration during the assessment of your application. Contextual factors may include: • Career breaks due to parental leave, or periods of illness. • Reduced time spent undertaking research due to a disability or caring responsibilities. This could include any physical or mental difficulty that may have impacted your research career. These are situations that have a significant impact on your ability to undertake research. • Reduced opportunities to career support e.g. mentorship, and limited opportunities to undertake prior research and training. • Impact of the COVID-19 pandemic on your research career. Please also use this section to detail any other factors that may have impacted your research career not listed in the examples provided. The impact on your career to date will be specific to your particular circumstances but could include such impacts as limited opportunities to obtain grant funding or fewer publications. In general terms, mitigating factors should be significant, and relevant. NIHR acknowledges that you may be reluctant, or uncomfortable disclosing relevant information that is sensitive. However, you should bear in mind that we are unable to take into account factors that you do not disclose. Please be assured that the information provided by you is sensitive and will be treated confidentially and in line with [General Data and Protection Regulations (GDPR)]().
1. Three periods of maternity leave and continuing parenteral responsibilities requiring part-time working and limited travel time/distance. 2. Specialist clinical role and limited opportunities for career progression or job mobility, financial and organisational constraints on staffing, recruitment and development of senior roles. 3. Dietetic positions at SGH are purely clinical roles. Any research and audit needs to fit around clinical responsibilities. I have mainly done this using my own unfunded time. I would not have completed the KRUK or current applications without additional TACRI funding and support of Prof Heather Jarman since 2023. I am the

only dietitian at St George's Hospital to have attempted this kind of application.

4. There have been limited mentoring or role models for non-medical research until recent advances and TACRI funded fellowship in 2023

5. Delay in making grant applications due to COVID-19 pandemic - high clinical demand and staff shortages - worked clinically throughout

4. Plain English Summary of Research

Title: Improving nutritional care for people with chronic kidney disease: a new way to identify and prevent malnutrition.

Background
Protein energy malnutrition develops when food intake does not meet the body's needs, or the body does not respond in the normal way. Malnutrition affects up to half of people with more advanced chronic kidney disease, including those on dialysis. This has many causes including poor appetite, fatigue, treatment side-effects and dietary restrictions. Malnutrition can be distressing, make people feel weak and tired, develop medical problems and serious illness, and spend more time in hospital. As with cancer, looking for malnutrition earlier makes it easier to prevent and treat. It also reduces healthcare costs.

Nutrition screening is the recommended way to look for malnutrition, guide care and decide whether to refer to a dietitian. To do this, nurses need the right 'tool', a standard set of questions leading to a care plan. General screening tools do not work well. Earlier attempts at kidney-friendly ones were too complicated and not tested in real-life settings. Patients and nurses were not asked for their opinions or to help develop them.

We previously developed and tested a screening tool for kidney wards. We will work with nurses and people with kidney disease to adapt it for outpatient services. This will include developing training resources and self-management information and testing these in three hospitals.

Aim

To work with nurses and people living with kidney disease to review and adapt an existing inpatient nutrition screening tool for outpatient services.

Methods
We will ask nurses and people living with kidney disease to take part. They will be from three large kidney units that serve communities with a range of social and ethnic backgrounds.

Study 1. Research review

To review what is known about nurse-led nutrition screening in kidney outpatient services, the tools used and views of nurses and service users. This will identify discussion topics and priorities for the codesign study.

Study 2. Codesign

To adapt the inpatient screening tool for outpatient settings with nurses and people with kidney disease. This will include planning how it is used and designing resources for training and self-management through several rounds of individual and group meetings.

Study 3. How it works in practice

We will introduce the new screening tool into specialist kidney outpatient services for people with advanced kidney disease and on dialysis.

We will check that the new screening tool works well in practice, using a recommended method. Nurses will complete it with patients as usual and give them some self-help information. Some patients will be asked the same questions by another staff member to check this gives the same result. For 139 patients who agree to take part, a kidney dietitian will do a detailed assessment and test handgrip strength to check for malnutrition and compare with the screening tool. Participants will be asked to use and rate the resources. We will ask participants and nurses for their opinions using a questionnaire. We will also assess whether having the screening tool helps nurses to include nutrition in care plans, give information and refer to dietitians.

Conclusion

This project fits with NHS goals for better patient care and quality of life and less need for hospital treatment and admission. It will provide the first practical, tested specialist nutrition screening tool for kidney outpatient services, designed with nurses and people with kidney disease. This will help to improve access to nutritional care for more people who need it. This includes future use of hospital electronic systems and online platforms. It will also help to improve future healthcare and malnutrition research.

5. Scientific Abstract

Background

Protein energy malnutrition affects up to 50% people with chronic kidney disease (CKD). Risk of malnutrition increases with disease progression. This can reduce quality of life, mobility and independence and cause poor wound healing and increased frailty, complications, ill health and time in hospital. Guidelines recommend nutrition screening for timely detection and treatment of malnutrition for patient benefit and to reduce costs. However, this is widely missing in renal services due to a lack of suitable screening tools. Therefore, malnutrition goes unrecognised, without intervention or referral to dietitians. Nurses have responsibility for nutrition screening but little involvement in screening tool development and the patient perspective has been lacking.

We previously developed the only nurse-led validated nutrition screening tool for specialist renal wards. It is cited in guidelines and used across the UK and worldwide. Adaptation for outpatients using codesign would provide a practical, consistent evidence-based approach to nutrition screening for renal units, and emphasise the importance of involving nurses and people living with CKD.

Aim: To determine if a codesigned adaptation of a nutrition tool, training and self-management resources is feasible, acceptable and valid for use within outpatient services for people with advanced CKD (stages 4-5) in three large renal units

Method:

There are three studies within the proposal:

1: A scoping methodology will be used to review the barriers and facilitators for delivering nurse-led nutrition screening within renal outpatient services and the characteristics of existing tools, with an emphasis on the nurse and patient perspective.

2: Codesign

A multi-stage accelerated form of experienced -based codesign will involve group and individual meetings with renal nurses and people living with CKD. There is a need to establish appropriate terminology, adapt the ward-based screening tool, design training and self-management resources.

3.Validation and Implementation

The codesigned nutrition screening tool and nurse-led screening will be implemented in selected renal outpatient services.

A multicentre validation of the screening tool will be conducted in 139 adult consenting participants with advanced CKD (eGFR <20), powered according to guidance for testing reliability and validity of a new or adapted instrument.

This includes testing concurrent, construct and face validity, sensitivity, specificity. The nurse-led tool and a patient self-screening version will be compared against a comprehensive extended dietetic assessment, using the validated subjective global assessment as the reference method, and against handgrip strength. A generic Malnutrition Universal Screening Tool (MUST) will be completed for comparison. Inter-rater reliability will be tested in a subsample of 52 consenting participants. Recorded data will include indices of co-morbidity, frailty and social deprivation, communication barriers and demographics. Participant and nurse feedback will be obtained by questionnaire.

Conclusion

A valid screening tool for CKD outpatient services, codesigned with people with CKD and specialist renal nurses, offers a new way of integrating screening for malnutrition at an earlier stage and in collaboration with the key people involved. It provides opportunity for earlier detection of malnutrition risk, better cost-effectiveness and more efficient use of dietitians. This will improve patient care and support technological and research advances.

6. Detailed Research Plan

What is the problem being addressed?

Malnutrition has serious effects on well-being, health and healthcare costs and is a common consequence of chronic kidney disease (CKD) (Chan 2021) [Figure 1]. Malnutrition often goes undetected by clinical staff and there is limited access to kidney dietitians, resulting in inadequate and inequitable nutritional care for people with CKD (Mafrici et al 2022). Nutrition screening is important to identify nutrition risk, direct care and reduce healthcare costs (Elia 2015). However existing methods are not suitable for CKD outpatient services (Lawson et al 2012, Jackson et al 2019; Kosters et al 2020). There has been little recognition of nurses and people with CKD as central to nutrition screening and management of malnutrition.

Why is this research important?

Malnutrition is associated with increased morbidity, mortality, hospitalisation and reduced quality of life (Stratton et al 2018). Significant cost savings are associated with nutrition screening, provision of information and nutrition support interventions (Brown et al 2020; Elia 2015; Stratton et al 2018). Successful implementation of the screening tool will help to prevent malnutrition and allow for earlier, more effective intervention. This will improve distribution of existing resources, including nursing and dietetic staff, reduce disparities in services and limit the individual and economic costs of malnutrition.

This work aligns with priorities to improve services for patients, support self-management, and reduce pressure on renal services (NHS 2019; NICE 2012). UK and international guidelines recommend regular screening for malnutrition with a validated tool (Ikizler et al 2020; Wright et al 2019). This work will help services for people with CKD conform with the guidance. Malnutrition and nutrition screening have been identified as an NIHR national research priority for patients and clinicians (James Lind Alliance, www.jla.nihr.ac.uk). A validated tool will enable good quality research into malnutrition and nutritional intervention strategies, including self management, for people with CKD.

A nurse-led screening tool for CKD services should be developed in collaboration people living with chronic kidney disease (CKD) and the specialist nurses who support them. Co-design of resources, participation of key stakeholders and validation of the screening tool in routine clinical use are key strengths of this proposal.

Review of existing evidence

CKD and malnutrition

CKD is classified into 5 stages according to kidney function, with an increasing burden of disease in the more advanced Stages 4 and 5, with an estimated glomerular filtration rate (eGFR) of <30 and <15ml/min/1.73m2 respectively. Over 1.8 million people in the UK are living with CKD Stages 3-5, with approximately 70,000 receiving dialysis (UK Renal Registry 2023).

Protein energy malnutrition affects an estimated 28-54% people with kidney disease (Carrero et al 2018). The estimated annual healthcare costs associated with malnutrition are over £20 billion per year in England alone, with malnutrition associated with almost four times the cost of treating someone without malnutrition (Brown et al 2020; Elia 2015).

CKD is associated with high hospital usage and malnutrition has been shown to be an independent predictor of the duration of hospital stay and associated with increased mortality rates and hospitalisation costs (Borek et al 2017, Jackson et al 2019, Kerr et al 2012). Renal dietitians have the expertise to identify and treat malnutrition but are a limited resource and patient access varies widely (Mafrici 2021). However, detection of malnutrition in advanced CKD is difficult, due to symptoms such as fluid retention and sarcopaenia, exacerbated by comorbidities such as heart failure (Ikizler et al 2020; Wright et al 2019;). Patients find it challenging to define and recognise malnutrition in themselves and non-dietetic staff often lack training to do so (Bullock et al, 2021).

Nutrition Screening

Nutrition screening is the first step in the prevention and treatment of malnutrition to reduce the adverse effects and the burden and cost of treatment. It should be a rapid and way to identify requirement for monitoring and early intervention and is recommended for all healthcare settings. (NICE 2012, Stratton et al 2018). Screening generates a care plan depending on level of risk indicated. This may include first-line information or a dietetic referral for more detailed assessment and specialist treatment.

Renal nutrition guidelines emphasise the importance of nutrition screening (Wright et al 2019; Ikizler et al 2020). Nutrition screening tools (NST) provide a standard method but need to be reliable and valid for the clinical setting and patient population for which they are intended (Sim & Wright 2005; Green and James 2013). Renal-specific tools are often complex and not practical or valid in clinical use (Ikizler et al 2020; Wright et al 2019). This prevents compliance with guidelines.

Generic tools such as the Malnutrition Universal Screening Tool (MUST) are widely used and recommended. However, studies have demonstrated that MUST is not valid in renal settings and can miss malnutrition in at least half of those affected (Lawson et al 2012, Jackson et al 2019; Kosters et al 2020). Nursing staff have the main responsibility for nutrition screening but have had minimal involvement in the development of screening tools. Public involvement is largely absent. There is opportunity for self-screening and management with appropriate training and resources, particularly with the advent of online portals.

Developed using a staged approach, the iNUT is the only nurse-led nutrition screening tool for kidney wards, to be validated whilst in routine clinical use, and with the positive feedback of nurses (Jackson et al 2019). It consists of a brief assessment to indicate malnutrition risk, with a simple scoring system to guide clinical judgement and care, such as additional monitoring, support and referral to dietetic or other services [Figure 2]. It is cited in national guidelines and increasingly used in renal and other specialities in the UK, and also abroad (Wright et al, 2019; Mafrici et al 2021; Romano-Androni et al 2023).

Outpatient services for people with CKD 4-5 vary considerably in structure and staffing and whether at a main hospital or community or satellite site. Services are frequently led by nurses with specialist training and skills in renal nursing and dialysis. Dietetic input varies widely creating inequity in care within and between UK renal units, ranging from 14 to 250 per 100 patients with CKD 4-5, with 59% renal dietitians reporting an unsafe or excess workload (Mafrici et al 2022). Globally only half of renal centres have dietitians involved in nutritional care but other staff often lack essential training (Crowley et al 2019; Bullock et al 2021; Weise et al 2018). Nutrition screening with a validated tool provides valued support for non-dietetic staff (Jackson et al 2019).

The perspective of nurses and people living with CKD

For nurses, barriers to screening include competing priorities, training and education, and discrepancy between attitudes and practice (Greene et al 2013). This has been little examined in CKD services. A codesigned approach to the development of training materials and implement strategy is planned to explore and resolve barriers in specialist renal settings.

There has been limited research into nutrition screening from the perspective of health service users. A recent systematic review of studies, involving people living with and without a range of medical conditions, indicated general acceptability of the screening process but poor knowledge of malnutrition and poor self-perception of risk, reducing opportunity for self-management (Bullock et al, 2021). A paediatric study indicated potential for anxiety and feelings of judgement and that a lack of understanding of the purpose of screening, next steps, and benefit to the individual patient could limit acceptance of nutrition screening. The codesign approach aims to explore and address this.

Study sites

Recruitment will take place at three hospitals in South London and Surrey: King's College Hospital (KCH), St George's Hospital (SGH) and St Helier Hospital (SHH), serving over 3000 people on dialysis alone with opportunity for diverse participation (UK Renal Registry 2023). Our previous study participation was 43% Black, Asian, mixed and other ethnicities (BAME) from a catchment where BAME representation is between 50% (Lewisham) and 15% (Surrey) population and English Index of Multiple Deprivation (IMD) rankings from around 300 to 63 according to Census and Ministry of Housing data. (Jackson et al 2019). Sites also provide diversity in nurse outpatient services, staffing and access to dietitians (Mafrici 2021).

Research Question

Is a codesigned adaptation of a nutrition screening tool and its implementation acceptable, feasible and valid for use by nurses with people with advanced chronic kidney disease attending renal outpatient services?

Research plan summary

1. Scoping review

Aim: To understand the barriers and facilitators for delivering nurse-led nutrition screening for CKD outpatient services, the characteristics of existing tools and the perspective of nurses and people living with CKD

2. Co-design (multicentre, three hospitals)

Aim: To work with renal nurses and people with lived experience of CKD 4-5 and attending renal outpatient services to codesign an adaptation of an existing inpatient screening tool to monitor and prevent malnutrition in CKD, including training and self-management resources.

3. Validation (multicentre, three hospitals)

Aims
- Phased training and introduction of the codesigned screening tool, strategy and resources into selected renal outpatient services.
- To determine the validity, reliability and acceptability of a co-designed nurse-led screening tool and self-management resources in routine use for people with advanced CKD attending renal outpatient services

The codesigned screening tool and resources will be tested in a validation study in three hospital renal units. This will include testing for validity, reliability and acceptability of the screening tool in clinical practice.

STUDY 1. SYSTEMATIC SCOPING REVIEW

((Year 1 of 4)

Research Question: What are the barriers and facilitators for delivering nurse-led nutrition screening in renal outpatient services encountered by nurses and people living with CKD and what is their knowledge, experience and opinion of malnutrition and nutritional screening?

Objectives:
- To undertake a scoping review of the quantitative and qualitative evidence on barriers and facilitators for delivering nutrition screening in renal outpatient services
- To explore the characteristics of existing tools and to determine the extent that nurse, patient or other service user perspective is represented.

A scoping method was selected due to the heterogenous nature of the evidence (Peters et al 2015). It will evaluate the source of information and research methodologies used, the applicability to people with CKD and the diversity in representation of staff and service users. The results will identify gaps and priorities to inform themes for the co-designed staged adaptation of the inpatient screening tool and development of training and information resources with nurses and people with CKD. The scoping review results will additionally be used to refine the validation study and for the planning of future research studies.

The scoping review will be guided by a six-stage methodological framework, including consultation with people living with CKD (PPI) and nurses (Arskey and O'Malley 2005, Levac 2010). It will be conducted and reported according to the systematic method PRISMA extension for Scoping Reviews (PRISMA-ScR) checklist (Pollock 2021; Tricco et al 2018).

Method:
i. Systematic search across relevant electronic databases and other sources such as Open grey, NHS evidence, NICE, and major clinical trial registries. Additional sources will include reference lists, nurse and service user organisations and conferences reports.
ii. Database extraction and identification of relevant research studies against exclusion/inclusion criteria
iii. Review of included studies by two independent researchers with a third independent reviewer in case of initial disagreement
iv. Description and mapping of studies to identify and highlight research deficits and inform the co-design study and NST development and implementation plan
v. Software utilised: EndNote reference manager, JBI systematic review software

STUDY 2. CO-DESIGN OF NEW NUTRITION SCREENING TOOL, TRAINING, RESOURCES AND CARE PATHWAY

(Year 2 of 4)

Codesign refers to the participation of service users and other stakeholders in the development of services. There is limited evidence of co-design in nutrition and diet research, and particularly of its use in the development of nutrition screening approaches (Tay et al 2021). The anticipated outputs will include a clarification of the preferred language and terminology relating to malnutrition, an adapted screening tool which includes guidance on care planning according to nutritional risk, nurse training resources and information on malnutrition prevention and intervention for people living with advanced CKD, according to nutritional risk.

The co-design study will be based around a multi-stage accelerated form of experienced -based co-design involving renal nurses and people with lived experience of CKD. The process of co-design will draw on evidence relating to person-centred self-management support developed by Bridges Self-Management (Bridges) and evaluated in other conditions. This approach is underpinned by co-design methods. (Jones et al 2016).

Objectives

A staged approach to codesign using groups and interviews with people living with CKD and renal nurses from three hospitals. The outputs of this stage will be determined by the co-design stages. Based on previous work there is a need to:

- establish appropriate terminology in relation to nutrition screening and malnutrition
- adapt an existing ward-based screening tool for clinic and dialysis settings
- design nurse training and care plan resources
- design patient self-management resources

The Process

1. Context mapping, drawing on the findings of the Scoping Review and mapping the current pathways of care
2. Interviews with people living with CKD and renal nurses. The findings will be summarised and discussed in the codesign meetings.
3. Codesign meetings for people living with CKD and renal nurses
4. Codesign of outputs by mixed groups of people living with CKD and renal nurses

Method

Purposeful sampling aims to create a heterogeneous group with diverse experiences. Sample sizes were chosen for active participation as recommended by Leask and colleague's framework for participatory co-creation (Leask et al, 2019). In case of drop out, effort will be made to recruit additional participants to maintain adequate representation. Drop out and additional recruitment will be reported. The Theoretical Domains Framework (TDF) guidance will be utilised (Atkins et al 2017). The co-design study will be supported by training, resources and facilitation from Bridges and guided by the Bridges Self-Management approach (Jones et al 2016).

Consent will be sought from all participants and expenses offered according to preference. Interviews and group facilitation will be led by the applicant. Bridges will provide training, tailored coaching and facilitation support to ensure quality standards and frameworks are met. Groups and interviews will be recorded, transcribed and summarised for use at the co-design meetings. Following these, interview data will be thematically analysed if indicated (Braun & Clarke 2006).

Software: otter.ai for group and interview transcription. NVIVO software for qualitative data analysis

Participants

Nurses

Inclusion criteria:

- Registered Nurses (RN) Band 5 or above
- Experience of working in renal outpatient services

Purposeful sampling, aiming for 10-12 participants with diversity in hospital site, experience (years), banding, gender and ethnicity. Reimbursement offered and participation certificate provided.

People living with CKD

Inclusion Criteria:

- Adults (>18 years) with experience of outpatient services for people with CKD 4-5, as a patient or carer
- Purposeful sampling, aiming for 10-12 participants with diversity in age, ethnicity and lived experience of CKD and malnutrition. Reimbursement offered according to preference
- Translation and interpretation will allow for up to 2 (10-20%) service-users with non-English preference to participate independently which is judged adequate in relation to reported local spoken English proficiency of 96.5-99% and discussion with PPI (Census 2021). Audio/large print materials will be available.

Group structure

Codesign groups held at each hospital site with a virtual participation option to be inclusive to those with preferences for in-person or online participation. Support and training with online access will be offered. Meeting length will be 1-2 hours depending on group numbers/ participant preferences, agreed in advance.

Stage 1: Separate small groups for people living with CKD and nurses to allow free discussion of experiences and issues.

For participants unwilling or unable to take part in the group meetings, individual sessions by phone or online will be offered. In advance, participants will be given information on the co-design process, discussion guide and resources to facilitate discussion.

Stage 2: One mixed (large) co-design meeting of 20-24 participants (2 hours).

To review the adapted screening tool and agree priorities and scope of the training materials and resources. A summary of the Stage 1 results and discussion plan will be sent prior to this meeting.

Stage 3: Co-design of training and self-management information.

Two co-design groups of 6-8 participants per group recruited from the total. To develop either a training and implementation package for nurses or self -management information for people living with CKD. Up to three online meetings will be held for each of these groups. Participants without online capability will be offered individual contact by phone or email for their input. Drafts will be circulated, and feedback collected by virtual circulation of drafts or online questionnaires (Microsoft Forms) where possible or by inperson, mail or phone contact.

Stage 4. Mixed (large) co-design meeting for final feedback and launch (20-24 participants)

STUDY 3. VALIDATION

Stage 1. Introduction of the outpatient screening tool into pragmatic settings (selected outpatient clinics/dialysis units), with training and ongoing support (1-3 months). There will be a site visit and half -day observation per clinic setting which will include seeking feedback from patients and the multidisciplinary team (as per codesign output). Within each setting the following will be monitored through standardised electronic database documentation.

a. Nurse-led dietitian referral rates
b. Presence of a documented malnutrition risk assessment
c. Presence of a documented nutrition management plan, dietetic referral or any other relevant intervention including other MDT referral.

Stage 2. Validation Study

This final stage aims to establish the psychometric properties of the new tool and its acceptability to service users and staff.

The objectives are to;

1. Explore construct and concurrent validity of the new tool against the current gold standard and related

measures.
2. Estimate inter-rater reliability (IRR) of the oNUT
3. Explore the predictive validity of oNUT in predicating admission rates and length of stay 6 months after assessment.
4. Provide evidence of face validity through eliciting nurse and participation opinion of the feasibility, acceptability, time taken to complete and relevance of the ONUT through a questionnaire survey.
5. Estimate the malnutrition rates by SGA in each setting and the total population.

Methods

Recruitment and participants

Using an online random number generator, a randomly selected sample of people from the clinic or dialysis list will be contacted at least one week in advance and offered verbal and written information about the study. During the clinic they will be approached for consent to;
1. a dietetic assessment by an experienced renal dietitian
2. completion of MUST and feedback questionnaires
3. collection of electronic record data including the o oNUT results and demographic information.

Translated information will be provided in up to three languages. Recruitment rates including non-attendance and non-consent will be reported.

Main inclusion criteria: 1. eGFR <30ml/min 1.73m2) 2. Attending an in-person specialist renal clinic or dialysis service, 3. aged >18 years. 4. All genders 4. Written informed consent.

Main exclusion criteria 1. No nurse-led screening tool completed at the clinic visit, 2. Not able to be assessed by the study dietitian within 1 week of the clinic visit. 3. Cannot be assessed by the reference method due to communication or other barriers. 4. Pregnancy (dietetic referral required regardless of nutritional status)

Sample size

The outpatient NST will be tested for substantial agreement with the reference standard (SGA by experienced renal dietitian) and a conservative estimation that 25% of sample are considered at risk of malnutrition based on the literature and local data (Carrero et al 2018). To be able to detect kappa=>0.7 (substantial agreement) as statistically significantly different from moderate agreement, kappa=0.4, with 90% power at a significance level of 5% and assuming proportion of positive ratings of 25%, 125 patients are required (Sim & Wright 2005). This has been increased to 139 to allow for 10% missing data. For the IRR analysis, to be able to detect kappa=>0.5 (moderate agreement) as statistically significantly different from no agreement above the role of chance, kappa=0, with 90% power at a 5% significance level, 43 patients are required and increased to 52 to allow for 20% missing data based on the experience of the inpatient screening tool research study (Jackson et al 2019; Sim & Wright 2005).

Statistical analysis

For continuous variables, descriptive statistics will include the number of observations, arithmetic mean, standard deviation, minimum, Q1, median, Q3, and maximum. For categorical variables, absolute and relative frequencies will be used to summarise the results.

Construct and concurrent validity of the oNUT will be estimated by the following statistics: sensitivity, specificity, negative predictive value (NPV), positive predictive value (PPV), kappa and area under the receiver operating curve; for a series of comparisons between the oNUT and SGA, MUST, and HSG. Results will be compared with the recommended target for sensitivity and specificity of above 80%, with 50- 80% rated ‘fair’ and less than 50% as ‘poor’ (van Bokhorst-de van der Schueren et al 2014). To estimate IRR of the oNUT weighted kappa will be used. Predictive validity will be analysed using regression models to predict admissions or length of stay using oNUT malnutrition risk while controlling for confounding variables. All estimates will be reported with 95% confidence intervals.

To test face validity questionnaires will include four-point Likert Score scales (to reduce central bias) and comments will be positive and negatively rated to allow for quantitative and qualitative analysis. Internal consistency will be assessed using Cronbach’s Alpha.

Data Collection Procedure

The adapted nurse-led nutrition screening tool (oNUT) will be in use at three hospitals as part of routine care in selected services for people with CKD4-5. Participants will be assessed during clinic or dialysis attendance (or at a mutually agreed time) within no more than one week of the routine completion of the outpatient screening tool by a nurse as part of standard care.

Participant assessments will be as follows:

1. Malnutrition Universal Screening Tool (MUST) completed for each participant in advance of the SGA, blinded to the result of the oNUT.
2. Subjective Global Assessment (SGA) according to the method of Detsky and colleagues and undertaken by a trained experienced renal dietitian (the applicant), will be used as the reference standard (Detsky et al 1987). SGA combines a clinical dietetic interview and examination, takes about 15-20 minutes. SGA is a recommended, validated method to categorize malnutrition status in renal patients (Ikizler et al 2020, Wright et al 2019). The SGA and dietetic assessment will be blinded to the result of the oNUT.
3. Handgrip Strength (HGS) using a hand dynamometer will act as an independent reference measure to assess construct validity. HGS is an objective non-invasive functional test of muscle strength and sarcopenia assessment, which correlates well with lean body mass and has prognostic value in renal patients (Ikizler et al 2020). HGS can be done sitting or standing, takes about 5 minutes using the maximum value of up to three attempts (participant-led) from the dominant or, if applicable, cannula/fistula-free arm. They will be categorized as malnourished by maximum HGS compared with European and UK male/female standards, and guidance for other genders (Dodds et al 2014; Linsenmeyer et al 2022).
4. All consenting patients will be asked to rate their nutritional status with the aid of the codesigned resource. The results will be compared to the dietetic and nursing assessments. Patient and care-giver feedback on acceptance, relevance and impact of the resources will be recorded.
5. Face validity is a subjective assessment of the nutrition screening tool for clarity, comprehensibility, and appropriateness for the target-group. Face validity of the oNUT in clinical practice will be evaluated through questionnaire survey of acceptability, time spent and relevance, distributed to staff and consenting participants. This is suitable to complete with a BSc/MSc student or staff (nurse, medical or AHP) desiring experience in research under supervision/mentorship of the applicant, as achieved in a previous study (Jackson et al 2019). Questionnaires will be designed with PPI input and refined according to the codesign. Questionnaires will be distributed and returned at clinics with the option of anonymous return to reduce bias. Contact details for participants will be collected separately for those who wish to be kept informed.
6. Participants will be asked to report education status, preferred language(s), literacy barriers and IT literacy (use of smart phone or online banking), access to internet.
7. Clinical and demographic data from patient records. These will include weight, height and estimated dry weight from routine clinic assessment, Clinical Frailty Score, CKD cause and longevity, age, ethnicity and gender. Charlson Co-morbity score will be calculated (Charlson et al 1987). IMD rating will be calculated from postcode (Ministry of Housing CLG 2019) and reported for the total population an indicator of socioeconomic status. Hospital admission rates and length of stay 6 months after oNUT assessment will be collected.

Dissemination, outputs and anticipated impact

See Figure 3 Logic Model

The results and outputs will be highly translational for national and international renal and other services. Results will be shared with renal and dietetic professionals and people with CKD through lay and scientific publications, social media, websites and meetings locally, regionally and internationally. This will be achieved through already established contacts. UK and international conference abstracts will be presented. A minimum of three publications in peer-reviewed journals will be produced.

Anticipated impact is for renal units to introduce the oNUT with the codesigned training and self-management resources. The resources will be translated and further evaluated with local collaborations as with the translated Spanish form of the iNUT (Romano-Andrioni et al 2023). Further collaborations are planned with national and international colleagues in nephrology and other specialisms such as liver and heart-failure. In future we aim to establish a website for other researchers and clinicians to access the screening tools and resources, register their projects and build collaborative clinical research. Through industry, health service, clinical and academic collaborations, we aim to integrate the screening tool and resources with electronic records and patient platforms.

Project management

Progress will be regularly assessed by PhD milestones and 6 monthly reports throughout and transfer from the MPhil to PhD expected within 12 months of enrolment by written report and oral presentation. The graduate student office requires six monthly progress reports against agreed milestones. Meetings with academic supervisors will take place every 1-4 weeks in-person or virtually with additional support as needed. SGUL postgraduate student guidelines will be adhered to, including attending all relevant local training and maintaining meeting and progress reports. A detailed project plan and timeline will be reviewed regularly to track progress and indicate if extra support is needed.

Ethics

Ethical approval will be obtained for the phase 2 codesign study and the validation stage of the phase 3 CKD oNUT feasibility study at each study site. The protocol, PIS/consent form and other study related documentation will be reviewed and approved by the Research Sponsor and Research Ethics Committee (Health Research Authority). CKD oNUT will be registered on an appropriate clinical trial repository.

Success criteria

Phase 1. Completion of scoping review

Phase 2.

- Recruitment of 10 nurses and 10 service users for codesign
- Completion of a co-designed adapted nurse-led nutrition screening tool, nurse and patient resources

Phase 3. Feasibility

- Introduction of oNUT to a minimum of one renal outpatient service at each hospital site, including onsite training and support with co-designed training package and service user information.
- Participant recruitment to target
- Feedback questionnaires return rate of at least 50%

Overall

- Project training and development plan completed
- Participation of at least one student or staff mentored/supervised by applicant
- Results disseminated, presented at UK and international conference and published in a peer review journal
- Thesis write up completed
- Application for advanced fellowship

Key Risks

1. Failure to recruit people with CKD. This is mitigated by recruiting via three large research active renal units and established relationships with clinical services. We will follow PPI advice on study information and protocols and feedback on clinical and research methods. There are hybrid and telephone options for the codesign groups and individual interviews for people who are unable to attend group sessions. Translation/interpretation will be provided to enhance participation.

2. Failure to recruit nurses. I have established interest and commitment at all sites and support from nurse and unit leadership, building on previous work, to mitigate this risk.

3. Failure to implement the screening tool. Established collaborations and contacts at all sites and support from nurse and medical leadership aims to mitigate this risk.

4. Failure to complete assessments within the clinic settings. The applicant has the experience and skills to work with in these clinical settings in liaison with the clinical teams.

7. Patient & Public Involvement

Please describe how patients/service users, carers and the public have been involved in developing this proposal

The research priority has developed through extensive clinical contact with patients presenting with malnutrition and limited self-management options. It is informed by previous study feedback, individual patient feedback and consultations with patient organisation.

We have planned this work following a previous study in the same three hospitals chosen to ensure diverse participation in social and ethnic backgrounds and experiences of kidney disease and services. Feedback from the 140 patients involved helped to plan this study, including the emphasis on patient perspective. We found a positive attitude to screening from a dialysis unit survey, with 78% reporting 'a regular nutrition check will help me get the care I need' and 67% that 'I would like to know more about spotting the signs of malnutrition'

We asked for the views of patients, representatives from the local Kidney Patients Association (KPA) and the kidney charities. Feedback has been incorporated in the research design and description. People find it difficult to notice signs of malnutrition. I identified a lack of information on malnutrition and will be working with Kidney Care UK (KCUK) and the Renal Nutrition Group of the British Dietetic Association to correct this.

There has been extensive PPI review of an earlier version of this proposal through the Kidney Research UK (KRUK) grant review process. This included the initial review of the full proposal, interview process and panel comments. In response I have revised the research plan to upgrade the codesign study, clarify the importance of nutrition screening and revise the lay summary. Since then I have had additional NIHR and KRUK PPI group feedback and further refined the plain english summary and research plan. I have tried to understand and respond to the challenges in communicating the importance of malnutrition and nutrition screening. One reviewer said there was 'no malnutrition in the UK'. Another thought it was mainly vitamin deficiency and due to poverty. However, a representative with personal experience of malnutrition recognised the impact on her health, and valued appropriate intervention and information. The range of understanding and experiences has helped design and structure the codesign and feasibility studies.

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Please describe the ways in which patients/service users, carers and the public will be actively involved in the proposed research, including any training and support provided

This proposal aims to be innovative in nutrition screening development by including the perspective of people with kidney disease and through the emphasis of their involvement in codesign. As such, PPI will be included throughout with the local PPI programme and nationally in association with Kidney Research UK and Kidney Care UK charities. This will included review of protocols, information and questionnaires.

Existing contacts and collaborations with patient charities such as Kidney Care UK will help disseminate findings through meetings, written information, websites, podcasts and social media. There is an established patient involvement programme and we will be working with the St George's Kidney Patient Association..

The recruitment sites have been chosen as they provide diversity including age, social deprivation, educational background, ethnicity and gender. Social deprivation, educational attainment, IT access, literacy and communication barriers will be evaluated in participants throughout and recorded as a barrier to participation, if applicable, to inform future research into nutrition screening and self-screening.

A scoping review has been chosen to examine the current evidence on the patient perspective of nutritional screening. Scoping review guidance emphasises stakeholder consultation. PPI before, during and following the review will include collaboration with planning, the conclusions, dissemination and ensuring a broad range of sources including patient publications and reports.

The emphasis on co-design in this project is novel in nutrition screening tool (NST) development, to ensure the active involvement of service-users in developing the tool, nurse training and self-management resources. Reimbursement will be given for participation and expenses. Purposeful sampling aims for a range of age, gender, CKD experience and BAME backgrounds. Audio and large print options for all written information and translation/interpreters to include non-English speakers will be provided if needed. Individual interviews will be offered to those who unable to participate fully in the group sessions, in-person or telephone options offered for those without online access. Training and support for online access will be offered. The participant perspective

form part of the analysis and reporting. This will be used to design and implement the nurse-led NST and design and test the patient self-screening tool.

If it is considered not appropriate and meaningful to actively involve patients/service users, carers and the public in your proposed research, please justify why
N/A

8. Training & Development and Research Support

Proposed training and development programme

The aims of this academic and clinical training and development programme are to:

1. Acquire the skills in research design, implementation and analysis to undertake research at doctoral and post-doctoral level.
2. Develop skills in scoping review, qualitative research and co-design.
3. Further my career as a clinical academic.
4. Be able to independently mentor and supervise dietetic and other students up to and including PhD level.
5. To be able to provide a higher level of mentoring and leadership in research for dietitians and non-medical health professionals (NMPs) locally, nationally and internationally.
6. To maintain and develop my advanced clinical skills.
7. To promote, support and advance the dietetic profession and other NMPs in quantitative and qualitative clinical research

Structured learning

PhD in Clinical Nutrition St George's University of London (SGUL)

St George's is the only UK university to share a campus with a teaching hospital and there is a close relationship between clinical academics and academics from diverse backgrounds. There is a comprehensive training program to provide me with the training to undertake research at doctoral level and beyond. My supervisory team will provide ongoing training and development support.

Year 1

SGUL Graduate Skills Training Program for PhD students: One day per week for 14 weeks, 9am-5pm

Comprehensive postgraduate foundation training programme Includes weekly sessions on Research Methods, Research Project Planning and Management, Statistics and Practical Data Analysis, CGP, scientific writing, systematic review, quantitative, qualitative and mixed methodology.

Cardiff University JBI Scoping Review one day workshop (online)

To understand theories and concepts relating to scoping reviews and other types of evidence synthesis. To support planning, execution and reporting of the scoping review following the Joanna Biggs Institute approach

An Introduction to Qualitative Research Methods 4 day course

Qualitative Health Research Centre (QUAHRC) at King's College London.

Includes interviews, groups, observation, ethical issues, and thematic analysis, NVivo. This will help prepare the co-design phase of the study and ensure I have a solid background for future work.

Public Involvement in Research

Imperial College London (Coursera) Four weeks x 4 hours (free): To support optimum public involvement for all phases and for future research.

Getting started with SPSS

Open University Free course (3 hours online): To refresh and update previous experience with SPSS.

NIHR ARC South London Applied Research Leadership Academy Jan-June (1 hour per week - free)

To

- Develop my research leadership qualities and the skills to design, and deliver research projects and ensure impact

- Understand how to strategically lead research at local/national/international levels and how to organise, give and receive mentorship/coaching

Year 2

Point of Care Foundation: Co-design Workshop

To provide foundation of codesign, prepare me for the specialist Bridges training and is essential for the co-design study and future aspirations in codesign research.

Bridges Training

To support the co-design phase and enhance my research, teaching and clinical skills and future aspirations. Co-delivered by people with lived experience of chronic disease. Includes:

1. Narrative Interview Training: Bespoke highly contextualised training including facilitated learning, practise opportunities and examples of good practice.
2. Facilitation Training: Bespoke workshop style training including facilitated learning, practise opportunities and examples of good practice. Will build knowledge, skills and confidence in facilitation skills for both online and in-person sessions.
3. Bridges Supported Self-Management Training: Open course eight hours of training fully contextualised to people living with long term conditions.

Year 3

ARC South London

1. Introduction to Implementation Science: 6 week online course

To understand the core principles and methodologies with training and resources for the planning, development, implementation and evaluation of health and care interventions. It will consolidate my previous learning and prepare me to lead the implementation of nutrition screening.

2. Implementation Science Masterclass

To advance my understanding and skills in implementation science. To support the current research proposal and future implementation of nutrition screening practices nationally and internationally.

The power of infographics in research dissemination (Open University - Free)

To be able to effectively utilise and evaluate infographics to present and communicate research.

Research Visits

Brisbane, Australia (and regional renal units)

The Australian and New Zealand Society of Nephrology annual conference

Ass Professor Helen MacLaughlin (secondary supervisor)

Queensland University of Technology, University of Queensland

Renal dietetic departments in Brisbane and the region

To consolidate learning and explore opportunities to collaborate, share best practice and promote nutritional screening. Australia is one of the few countries with dietitians working in a similar capacity as in the UK and Brisbane is a centre for clinical academic renal dietitians. ANZSN represents over 1000 renal health professionals in Australia, New Zealand and the region.

Objectives

- To visit key renal units, including those serving less privileged communities, for exchange of knowledge,

resources, practices and information
- To network with global leaders in clinical renal dietetic research and explore opportunities for collaboration in clinical research
- To evaluate similarities and differences in renal nutritional assessment and screening between Australia and the UK
- To participate in lecturing/training for university and clinical staff and students in nutrition screening and renal dietetics
- To participate in the ANZSN annual conference
- To explore opportunities for iNUT and oNUT screening tool implementation and evaluation in international contexts

Other conferences to present research findings:

I am requesting a contribution for travel to attend (other costs will be funded separately):
1. European Society of Parenteral and Enteral Nutrition. To disseminate to European nutrition professionals.
2. UK Kidney Association (UKKA) Conference (multi-professional /service-user/industry/charities). To disseminate and discuss research with key stakeholders.

Clinical practice

I will use 20% of my time in dietetic care, service development, audit and policy. I will maintain skills in nutritional assessment including subjective global assessment and continue to develop my motivational interviewing and behavioural modification skills. I aim to utilise the Fellowship training to develop and evaluate group education and teaching.skills.

Enhancing Research

I will continue to lead and mentor dietitians, develop research collaborations and support specialist nutrition screening within the Trust and externally. I will provide mentoring and support for research grant applications to non-medical researchers and dietitians within the Trust and nationally, and participate in UKKA guidelines development.

Primary Doctoral Supervisor

PRIMARY DOCTORAL SUPERVISOR 1

Name	Fiona Jones

Institution	St George's, University of London

Position	Professor

Description of support provided
Professor Jones is ideally placed and committed to being primary supervisor on this award. She is a Professor of Rehabilitation research, founder/CEO of the social enterprise Bridges Self-Management and a physiotherapist by background. She has extensive experience of participatory methods, co-design and implementation science and principles of collaboration, relationships and cultures within healthcare. She has experience in scale development psychometric testing, and developed the stroke self-efficacy questionnaire which is used extensively in stroke rehabilitation and translated into more than 20 languages. Her research involves development of self-management approaches for people living with many different long-term conditions. She has just completed and NIHR funded study, LISTEN, using codesign in the development of a self-management intervention for Long-COVID. She has raised over £4million to support research in co-design and person-centred care and her research has informed National Clinical Guidelines for stroke (UK) (2016, 2022). She will provide content and methodological expertise to inform my Doctoral research. Professor Jones has published 98 peer-reviewed papers (H-index 22) and supervised 11 PhD students to completion. She is currently supervising 3 PhD candidates and mentors clinical academics (AHPs) at predoctoral, doctoral and fellowship level.

She is co-applicant on a grant for an NIHR hub to support community AHPs and nurses to become active researchers.

Professor Jones has supported the development of this application and we have developed an excellent working relationship. Throughout the DCAF we will meet at least twice monthly, in person or online, with additional, ongoing communication by phone, Teams and email as needed.

Additional Academic Supervisor(s)

ADDITIONAL ACADEMIC SUPERVISOR 1

Name	Helen MacLaughlin
Institution	Queensland University of Technology, Australia
Position	Associate Professor

Description of support provided

Prof MacLaughlin is an expert on renal dietetics and nutrition screening and a valuable part of the supervisory team for this DCAF application.

Prof MacLaughlin is currently supervising 3 PhD students, 2 are part-time and are both 4 years in, and the first is nearing completion. The 3rd is in her first year. She has supervised 3 Masters by research to completion and over 40 honours students. She is also supervising an NIHR Advanced Fellow..

As a clinical academic Prof MacLaughlin's strength is supervising pragmatic research undertaken in clinical settings, mostly in CKD populations. She has an established track record of working with me in research (3 joint papers) and collaborations professionally, BDA RNG national guidelines and local best practice sharing.

She has capacity to support me as a co-supervisor and is a topic expert in nutrition in CKD research with >30 publications in this field. We have demonstrated an effective supervision relationship in the work up of this application over the last 12 months, with regular meetings and email communication resulting in the successful submission of this application and a previous submission to KRUK which reached final interview stage.

She will facilitate my research visit to Brisbane with a full programme of meetings, introductions and research/clinical collaboration opportunities in year 4 and will support my aspiration for international impact.

Prof MacLaughlin and Prof Sue Green met on an NIHR leadership training program and were keen to take the opportunity of working together and with me as co-supervisors.

ADDITIONAL ACADEMIC SUPERVISOR 2

Name	Sue Green
Institution	University of Bournemouth
Position	Professor of Nursing

Description of support provided

Sue Green is a Professor and Deputy Head of the Department of Nursing Science. She is committed to the development of clinical academic roles for nurses and other non-medical professionals. As well as a Registered

Nurse she is also a Registered Nutritionist. Her research is focussed on nutritional care by nurses, nutrition screening and the effect on nutritional care. She was awarded Nutrition Nurse of the Year at the British Journal of Nursing Awards 2021. Professor Green and Prof MacLaughlin met through a NIHR Leadership training programme and are keen to work together and as co-supervisors with me for this proposal which ideally reflects their skills and experience. Prof Green has supported me over the past year in the development of an application for a Kidney Research UK Grant and through the current application. She provides essential nursing representation and expertise for this work that aims to represent the nurse perspective. We will communicate by email and monthly virtual meetings throughout the fellowship with additional communication by email. She has supervised a total of 8 PhD students with an 100% success rate.

Clinical Supervisor(s)

PRACTICE SUPERVISOR 1

Name	Professor Debasish Banerjee

Institution	St George’s University of London & St George's Hospital NHS Trust

Position	Consultant Nephrologist, Care Group Lead for Renal Services

Description of support provided
Professor Banerjee is an international expert in renal medicine and leads the St George’s Hospital Renal Unit and the plans for a regional research unit. We have had a close working relationship over many years, He is very supportive of the dietetic role and opportunities for personal development. He brings an essential medical and clinical academic perspective, honesty and critical thinking and excellence in teaching, winning many teaching awards including the ‘Best Undergraduates Medicine Teacher Award’ and the ‘Overall Contribution to Education Award’ from St George’s Hospital. He has received multiple research grants and has previously supported me in my successful application for a British Renal Society Research Grant Award and during the study and is co-author of the resulting publication (Jackson et al 2019). He has supported the recent successful application for a SGH/SGUL Translational and Clinical Research Institute Fellowship and a Kidney Research UK (KRUK) Fellowship application which reached interview stage. He has supported me in participating in UKKA guidelines development and as an invited conference speaker and inspired my interest in the care of people with multiple conditions. He is active in the International Society of Nephrology Action of Clinical Trials, has co-chaired the patient engagement group and has multiple international collaborations. I will draw on his experience. Prof Banerjee has supported me with this application, is encouraging me to be ambitious in my career aspirations as a clinical academic and his wide-ranging academic and clinical experience and expertise provide an excellent basis for practice supervision

PRACTICE SUPERVISOR 2

Name	Gemma Stott

Institution	St George's Hospital

Position	Consultant Dietitian, Professional Lead for Dietetics

Description of support provided
My role as Consultant Dietitian (since 2016) and Professional Lead is to provide clinical and managerial Leadership for the Dietetics Department and within the Trust. I have experience as Chief AHP and Digital Champion. Helena is a valued and respected member of the Dietetics Department and Renal Unit, particularly from her knowledge of renal dietetics, nutrition screening, and longstanding interest in research and audit. I have been working with non-medical professional research leads within the organisation to recognise the importance of dietitians in research and create opportunities for Helena and other dietitians. I have encouraged her in her leadership development such as leading our dietetics research and audit group and mentoring dietitians and students. We value the respect from her medical, nursing, dietetic and other professional colleagues within the renal unit at St George's, in the region and nationally, and her commitment to self-development and voluntary work with organisations such as UK Kidney Association and the BDA Renal Nutrition Group. Also, her ability to create working relationships with dietitians in the UK and abroad through hosting, mentoring and teaching work. I and her line manager have encouraged her to seek opportunities in research, including attending meetings and conferences and to raise the profile of dietetics in research. We look forward to continuing to support Helena whilst she undertakes the Fellowship. This includes her participation and support within the Dietetic Department, in clinical development, supervision, teaching and mentoring. I have no hesitation to support Helena in this application.

Collaborations

Dr Sarah White , Senior Lecturer in Medical Statistics, St George's, University of London - lectures for the SGUL Graduate Skills Training Program has provided advice on the research plan and has committed to provide individual support throughout the Fellowship, particularly for the qualitative statistical content. This includes training on scientific writing, support with statistical analysis and co-author on journal publications.

Eleri Wood, Consultant Nurse, King's College Hospital - leads the Advanced Kidney Care (pre-dialysis) and supportive care service alongside leading Nephrologists and is an expert in patient peer support. Eleri provides essential nursing expertise, a valuable role model, critical thinking and strong leadership. She will provide research and clinical site support. The reputation of KCH in renal and multi-disciplinary research will support the dissemination and impact of the work and provide opportunity for future collaborations.

Rochelle Blacklock, Clinical Lead Renal Dietitian King's College Hospital - is active in multidisciplinary research and supporting others. She has collaborated on several peer review publications. We have an established working relationship. She and her team will provide site clinical and research support and increase opportunity for dissemination and impact.

Dr Pauline Swift, Consultant Nephrologist, St Helier Hospital. Research Director at one of the largest UK renal units, leading on the new Regional Research Unit which will provide opportunity for my ambition to become a clinical renal nutrition academic and lead on codesign. She supports my research proposal and the clinical introduction of nutrition screening. She will assist ethical applications and provide site support, advice and access to relevant services, and enhance dissemination and impact of the study.

Kidney Care UK is a charity and innovative provider of dietary information. I have reviewed and written information, including initial content on malnutrition and will continue this throughout the fellowship.

Kidney Research UK is a charity supporting research in kidney disease and have supported PPI for this proposal with PPI volunteers and lay research advisors . We will continue this collaboration throughout the fellowship.

St George's Kidney Patients Association represent people living with kidney disease who are committed to 'improving the lives of kidney patients at the hospital'. They are actively involved in many aspects of the Renal Unit. They have provided advice and will support with PPI and codesign and dissemination.

The **UK Kidney Association** is the leading UK renal professional body. Membership includes multidisciplinary clinicians, academics, industry and charity representatives and people with lived experience of kidney disease. They have supported me in leadership training and volunteer opportunities and I will continue to seek opportunities for collaboration and development.

Industry partnerships.

Stanningley Pharma,. I will continue and develop my long term association with this UK renal nutrition company, such as sponsorship as an invited speaker or contributor to dietetic and multi-professional meetings and journals.

Patients Know Best (PKB) is the UKKA public data provider. I have had positive response to initial meetings to explore options for self-screening and information

Mentor:

Dr Lina Johansson, Honorary Lecturer (Imperial College London), Clinical Academic Renal Dietitian (Imperial College Healthcare NHS Trust)

Lina has been a role model and mentor to me for over 10 years and introduced me to codesign and her academic collaborators. She is one of the few post-doctoral clinical academic renal dietitians in the UK at one of the largest renal units. She has past success in NIHR grant applications. We have worked together on renal nutrition guidelines and a systematic review. She is extremely responsive, generous and insightful with her criticisms and advice. She has agreed to regular online meetings throughout my fellowship and we aim to collaborate on future codesign studies.

Host Organisation Support Statement

The following statement of support should be composed jointly by the Heads of Department / Senior Managers of the employing host organisation and the partner organisation.
The Heads of Department / Senior Managers of these organisations will be able to add the statement of support to the form once they have confirmed participation. Invitations to participate are sent by the applicant via the 'Participants and Signatories' section of the form.

Statement of support

We confirm as senior academic mentors at St. George's University Hospital NHS Foundation Trust (SGUHFT) and St George's University of London, that we are fully supportive of Helena's proposed research and application for an NIHR Doctoral Clinical Academic Fellowship.

As NHS host institution and clinical partner, SGUHFT is one of London's biggest hospital Trusts, providing services to 3.5 million people with a staff of over 9000. The Trust is committed to promoting equality, diversity and inclusion which is represented in the Trust values and through the NHS workforce disability equality standards. The senior leadership team in the Trust recognise the critical role that the allied health professions have in expanding research capacity, patient involvement, patient outcomes and experience. The Trust has developed a sustainable programme of training and research to increase this research capacity, literacy and capability; to this end the Trust research strategy 2023-2028 makes specific reference to investing in staff wishing to pursue clinical academic careers by funding training on research skills and methods, and in providing protected time for research.

The Translational and Clinical Research Institute (TACRI) hosted by St. George's University Hospitals NHS Foundation Trust with St George's University of London supports researchers of all levels engage in research and grow our academic and clinical partnerships. We have an established track record of supporting clinicians who wish to become research active. Regular research seminars showcase research by Trust staff and research training is provided through bespoke workshops which would support Helena in her Fellowship e.g. patient and public involvement in research, funding opportunities. The Trust has a doctoral and post-doctoral research forum of which Helena will be part and which exposes her to a wider network of allied health professionals who are also who are also developing and succeeding in clinical academic careers. Under this strategy the number of successful NMAHP research fellowships has grown with current NIHR PCAFs (x3), DCAFs (x4), Senior Practitioner Award (x1) and ACAF (x1) in the Trust.

Helena's PCAF will be supported by PHRI, a multidisciplinary research unit with an outstanding record of conducting high quality research to underpin population health improvements. PHRI comprises ~60 staff, with 19 principal investigators. PHRI research income has been raised from funding bodies, including UK-MRC, the Wellcome Trust, the European Union and UK Governmental agencies and, particularly, the NIHR, which has accounted for >50% of total research grant income in recent years. We have a strong history of supporting NIHR clinical academic allied health professionals and primary care practitioners. All institute academics devote 80% of their time to research.

PHRI is well placed support the underpinning research and co-design of a tool for screening for malnutrition of renal patients under the guidance of Professor Fiona Jones. PHRI also offers expertise in allied fields encompassing behaviour change intervention development, study design, statistics, public health, and rehabilitation. Helena will be accommodated within PHRI and supported by fortnightly supervision meetings, access to our early career research group and regular seminar series. Helena will access the SGUL Graduate School Skills Programme as appropriate. This provides key transferable skills mapped against the Vitae Researcher Development Framework, addressing research practice, integrity, ethics, time-management, public engagement and core disciplines (statistics, methods, planning, critical appraisal). PHRI has a strong record of supporting career progression; many staff (professors and lecturers) were originally clinical academic fellows or PhD students within PHRI.

SGUL is fully committed to Equality, Diversity and Inclusion employment practices, as emphasized in the SGUL 2030 Strategic Vision and overseen by the Dean for Equality, Diversity and Inclusion. We are committed to the VITAE Concordat to support the career development of researchers and to the principles of the European HR Excellence In Research Award (HREIRA). SGUL has held an Athena Swan Silver Award since 2018 and is a Stonewall Diversity Champion and Disability Confident Employer.

This application aligns with the SGUL 2030 Strategic Vision and the PHRI 2022-2030 Strategy, which includes influencing the health policy agenda in the UK and globally; increasing the involvement of the public and patients and use of co-design in our research activities; supporting our academics to develop their research profile, through recognition by external bodies and stakeholder groups; and focusing on delivering research with high quality impact. SGUL has ARC South London membership which further facilitates co-production of research through patient and public involvement in research.

Helena has demonstrated a long-standing commitment to academic practice and was awarded a competitive Trust-funded Senior Research Fellowship in 2023 in recognition of the strength of her potential as a future academic leader. She used the time awarded on the Fellowship to develop her DCAF application.

The Trust has an agreed pathway for the development of clinical academic posts at pre- and post-doctoral level, and in the medium term we aim to support Helena beyond a successful DCAF and PhD by developing a specific clinical academic role for her to apply for on completion of this fellowship. The renal team in which Helena is based has a strong academic track record and is a developing multi-professional research group of which Helena will be part. She will continue in her practice role for 20% of her time on the fellowship and remain fully integrated within the clinical team.

Professor Heather Jarman, Research Director for Nursing, Midwifery and Allied Health Professions St George's University Hospital NHS Foundation Trust

Professor Charlotte Clarke, Director, Population Health Research Institute, St George's University of London

9. Detailed Budget

The finance section should provide a breakdown of costs associated with undertaking the research as described in the proposal.

1

Staff type	Lead Applicant

Name	Helena Jackson

Employing organisation	St George's University Hospitals NHS Foundation Trust

Grade	Afc Review Body - Band 7

Starting point on scale	09

Increment Date	01/05/2026

Justification of Costs

The budget was compiled with Finance Advisor Oliver Palmer, my supervisory team and the Research Governance Team at St George's. It considers the research proposal and training and development programme with cost-effectiveness and efficient use of resources.

Salary & training and development

Helena is the lead applicant of this application. She will be dedicating 80% of her time to this project. She is on Afc Review Body Band 7 - 09.

The training and development program consists of my PhD studies, courses, workshops, a research visit and conferences to produce and disseminate high-quality research, develop key skills as a doctoral-level researcher and progress as a clinical academic.

There is an absence of dietetic representation in research and guidelines on the clinical care of people with CKD. This impacts on the quality and impact of the work. The DCAF will support me to take on leadership roles in non-medical and codesign research, in national and international clinical guideline groups and in the national and international evaluation and implementation of the screening tool and resources.

Anticipated economic benefits:

Economic benefits are anticipated through the improved use of dietitians, prevention and treatment of malnutrition. The cost of treating an individual with malnutrition is estimated at almost four times that of the cost without malnutrition (Brown et al 2020; Elia 2015). .

Anticipated benefits for individuals with CKD and renal services:

Nutrition screening enables earlier and more effective treatment of malnutrition to reduce the adverse effects which include distress to individuals and families, increased hospitalisation, delayed recovery from illness and surgery, depression, reduced quality of life, mobility and independence and other serious physical and mental effects.

There is limited and inequitable access to specialist renal dietitians (Mafrici 2021). Funding constraints and lack of staff availability and recruitment means that there needs to be more effective and efficient use of renal dietitians.

Nutrition screening by renal nurses would utilise specialist nurse expertise and existing routine and regular contact with service users. It would enhance and standardise care. A valid nutrition screening tool facilitates training in recognising malnutrition. The codesign element is vital to ensure the project provides the tools they need and will support patients, families and carers in taking an active role, with less dependency on staff to recognise any difficulties. It will help to increase the confidence of service users in raising issues of nutritional concern and access the help they need from dietitians and other members of the MDT. Feedback from renal nurses and dietitians is supportive of nutrition screening and there is widespread demand for a validated tool for both inpatient and outpatient services (Jackson et al 2019). The availability of a validated tool will also facilitate research in nutrition and CKD.

Benefits are expected on a local, regional, national and international basis due to the current lack of renal specialist screening tools. The application is not limited to renal services as the iNUT inpatient tool is already used by a Heart Failure Unit and is being evaluated for Liver Services.

Research costs

The focus on codesign will add significantly to the understanding of perspective of nurses and people with CKD in relation to malnutrition and prepare me with the skills and knowledge to lead and collaborate on codesign research in the future. It introduces what should be an essential part of nutrition screening tool development and I hope will be a benchmark for future studies.

The proposed study would be difficult to achieve without the skills and experience of the current Applicant. The validation stage requires an experienced renal dietitian trained in subjective global assessment and other assessment methods and who can respond appropriately to both clinical and research concerns from participants, including appropriate referral to the clinical team. The proposal additionally benefits from experience of the Applicant in nutrition screening development and evaluation and from clinical and research experience at all three sites and from multiple collaborative relationships in the UK and abroad. This includes the expert multidisciplinary supervisory team, other clinical academics and charities, professional organisations and a digital platform provider.

This project is in keeping with the NHS Long Term Plan for digital and technological advances to outpatient care and self-management is to deliver digitally-enabled care that provides fast and convenient support for patients. A validated screening tool will have the evidence needed for NHS Trusts to incorporate specialist screening in their electronic records as demonstrated by the iNUT study (Jackson et al 2019). For the charities, professional organisations and digital platforms, the PPI and codesign of the tool and resources will be essential for future adoption and futher dissemination.

Additional research costs have been calculated on the SoECAT but will not be incurred as events will be performed by the central research team (ie. the lead applicant will facilitate the focus groups for study 2 and undertake all the clinical assessments for study 3) and are not considered a research support cost. No excess treatment costs have been budgeted for study 2 and 3 as standard care will not be affected.

My proposal includes essential PPI engagement throughout and has been costed as per NIHR INVOLVE Guidelines. Final costs may be lower than the budget if the codesign participation rates areat the lower acceptable range or if charity volunteers decline payments. Final payments will be agreed with each individual and the charities prior to PPI work.

Details of Posts and Salaries Costs

	Total (£)
Salary (£)	50056
Geographical Weighting (£)	7746
Other allowances (£)	364
Superannuation (£)	8312
Nat. Insurance (£)	6721
Current Annual Costs (£)	**73199**

Direct Costs
These are the costs that are specific to the research, which will be charged as the amount actually spent and can be supported by an audit record.

Staff Type: (Lead Applicant)-Ms Helena Jackson
Type of cost: NHS
% Full-time on this research: 80.0
Total months on this research: 45.0
Justification of Cost: Helena is the lead application on this grant. Dedicating 80% of her time and is graded on Afc Review body Band 7 - 09

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
Basic Salary	42047	44149	46356	48674	181226
National Insurance	6982	7331	7698	8083	30094
Superannuation	5951	6249	6561	6889	25650
Geographical weighting / Other allowances (if applicable)	6507	6832	7174	7532	28045

Totals	Applicant	Support Staff	Shared Staff	Total Cost (£)
Total HEI	0	0	0	0
Total NHS	265015	0	0	265015
Total Commercial	0	0	0	0
Total Other	0	0	0	0

Travel, Subsistence and Conference Fees

Type of cost: NHS
Description: Journey Costs
Conference: True
Item Description: Travel to conference and visit to Supervisor
Justification of Cost: ANZSN and Brisbane Estimate flight cost (2024 price estimation) : £1400 return ticket to Brisbane plus £200 for 1 x internal return flight to conference Other conferences to present research findings: As requesting full costs for these conferences will exceed the £3000 cap SGH Dietetics and Renal Departments currently commit to pay registration fees for those who are presenting, I am requesting a contribution for travel only to attend • European Society of Parenteral and Enteral Nutrition £250 (estimated travel cost to European City from London). To disseminate to European nutrition professionals – global reach. Previous research on renal inpatient screening was published in the ESPEN Journal: Clinical Nutrition. • UK Kidney Week x3 (multi-professional including service user renal conference). organised by the UK Kidney Association –£200 (estimated travel cost from London to UK City) This is an important way to discuss research priorities and disseminate and discuss research findings with key stakeholders including industry representatives and members of national renal professional/service user groups and charities

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
	0	100	100	1950	2150

Type of cost: NHS
Description: Subsistence
Conference: True
Item Description: For 1 week accomodation in Brisbane and 3 days ANZSN conference accomodation
Justification of Cost: Estimated basic accommodation in Brisbane (based on Booking.com options) £50 per day x 10 days = £500)

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)

	0	0	0	500	500

Type of cost: NHS
Description: Conference Fees
Conference: True
Item Description: The Australian and New Zealand Society of Nephrology annual conference
Justification of Cost: Conference Registration (2024 prices, post-graduate student early bird) $630 Aus = £315 ANZSN is a not-for profit organisation representing the interests of over 1000 health professionals committed to the prevention and treatment of kidney disease. There is an Annual Scientific Meeting in August/September. Members support research, education and clinical care initiatives to promote evidenced based practice and quality outcomes for patients in Australia, New Zealand and the region. This will be an important route to communicate my research findings and form collaborations for further research including implementation of the screening tool and further research.

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
	0	0	0	315	315

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
Travel, Subsistence and Conference Fees Total	0	100	100	2765	2965

Equipment

Type of cost: NHS
VAT relief to be applied: False
Description: Handgrip dynamometer (Jamar electronic)
Price excluding VAT £: 400
Expected life time: 60
Justification of Cost: Required for the validity assessment of the outpatient nutrition screening tool (Study 3). Jamar is widely used in clinical practice and research. The digital version makes it easier to record an accurate result.

	Year 1 (exc VAT)	Year 2 (exc VAT)	Year 3 (exc VAT)	Year 4 (exc VAT)	Total Cost (£) (exc VAT)
	0	0	400	0	400

Type of cost: NHS
VAT relief to be applied: False
Description: Digital recording equipment
Price excluding VAT £: 128
Expected life time: 60
Justification of Cost: voice recorder (eg SONY ICD-PX333) and 32GB SD card to record with permission: PPI feedback, supervisor meetings and co-design groups

	Year 1 (exc VAT)	Year 2 (exc VAT)	Year 3 (exc VAT)	Year 4 (exc VAT)	Total Cost (£) (exc VAT)
	128	0	0	0	128

Type of cost: NHS
VAT relief to be applied: False
Description: Laptop
Price excluding VAT £: 500
Expected life time: 45
Justification of Cost: A laptop of a minimum of Intel i7 and RAM 8GB is required for data collection, analysis and dissemination, home-working and multicentre working. It will need appropriate encryption, high memory capacity, remote hospital profile and ability to run SPSS, NVIVO and other relevant software.

	Year 1 (exc VAT)	Year 2 (exc VAT)	Year 3 (exc VAT)	Year 4 (exc VAT)	Total Cost (£) (exc VAT)
	500	0	0	0	500

Type of cost: NHS
VAT relief to be applied: False
Description: Tablet and protective case with handle
Price excluding VAT £: 300
Expected life time: 45
Justification of Cost: For electronic questionnaire completion for codesign groups and during feasibility study (to reduce use of paper and to improve efficiency of data collection, where possible). Current price for a iPad 10.9 (2022) 64GB from Backmarket (refurbished with 24month warranty) plus protective case from Amazon (-VAT)

	Year 1 (exc VAT)	Year 2 (exc VAT)	Year 3 (exc VAT)	Year 4 (exc VAT)	Total Cost (£) (exc VAT)
	0	300	0	0	300

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
Equipment Total	628	300	400	0	1328

Consumables

Description: Printing conference posters x 5
Type of cost: NHS
Please describe fully: Printing conference posters x 5
Justification of Cost: Aiming to submit abstracts of research for oral or poster presentation to UKKW conference x3, ESPEN x 1, ANZSN x1. If possible I will re-use posters depending on requirements of each conference Each poster will also be presented at local hospital research meetings and regional UK meetings.

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
	0	40	40	120	200

Description: Printing, photocopying and postage
Type of cost: NHS
Please describe fully: Printing, photocopying and postage
Justification of Cost: Bulk printing of participant information & consent forms, paper documentation for codesign study, PPI information and questionnaires where participants have a preference for non-electronic formats/mail SGUL inhouse bulk printing costs are £50 per 500 double sided black/white and £150 for colour. Colour will only be used where deemed necessary by the codesign groups, eg for training/self-management resource development. Virtual documentation will be used where possible for cost and environmental reasons.

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
	200	350	350	50	950

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
Consumables Total	200	390	390	170	1150

Patient and Public Involvement

Description: Translation and interpreters
Type of cost: NHS
Please describe fully: Translation and interpreters
Justification of Cost: To provide inclusivity to non-English speakers a budget for translation and interpreters is included for the Codesign (Study 2) and Feasibility Study (Study 3). For Study 2 and 3 translation in up to three languages (for one per site) is budgeted for at £130 (+VAT) per 1000 words (Total £1111.50) To include for each study patient information sheets (est 2000), consent forms (est 250) and plain language reports (est 600) based on estimations from www.atlas-translations.co.uk Translated information, where relevant (eg plain language report) will be utilised for dissemination of the research. Two interpreters are budgeted for seven hours participation in study 2 codesign groups (to allow up to 20% participation from non-English speakers) at £23 per hour from Language Line (£322). For Study 3 the estimation of £483 includes translation for up to 14 participants (10% of the target sample of 139) for the initial contact telephone call (30 mins) and clinical assessment (60 minutes). Costs for translations and interpreters are split across corresponding years according to the study schedules detailed on my Gantt chart.

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
	0	878	1039	0	1917

Description: PPI costs
Type of cost: NHS
Please describe fully: PPI costs
Justification of Cost: PPI is essential throughout the project,, including the scoping review planning, discussion and conclusions and dissemination, Reviewing documents; information sheets and consent forms for interviews, groups and the feasibility study. Reading and participating with publications. For PPI participants costs and reimbursement will be offered for individuals as per INVOLVE rates but take account of individual preferences and welfare benefit circumstances. For charity volunteers, reimbursement is calculated in the same way but final costs will be given according to charity guidance. Costs have been estimated based on 3 online meetings per year (including year 4) of 4 people per meeting at a cost of £55 per participant (including £5 for online expenses and allowing time for review and response times. I plan to request participation in the scoping review publication so will additionally request £50 per participant for up to 4 participants (£200).

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
	860	660	660	660	2840

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
Patient and Public Involvement Total	860	1538	1699	660	4757

Other Direct Costs

Description: Access to Bridges coproduction and codesign resources
Type of cost: NHS
Please describe fully: Access to Bridges coproduction and codesign resources
Justification of Cost: Access to a suite of tools that will be regularly updated and added to help with co-production and co-design in your work for the duration of the project. This toolkit contains resources and tools that have been designed by Bridges and others in the field. Includes detailed description and instructions for tailored use. These will be used for this and future research/clinical work

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
	0	200	0	0	200

Description: Bridges support in developing project resources
Type of cost: NHS
Please describe fully: Bridges support in developing project resources
Justification of Cost: Support from Bridges Self Management to identify resources to facilitate the co-design sessions, engagement resources, communication materials, and support in developing the project outputs. Up to 16 hours support Will ensure high quality codesigned outputs are delivered

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
	0	1600	0	0	1600

Description: Co-facilitation at each of the chosen sites
Type of cost: NHS
Please describe fully: Co-facilitation at each of the chosen sites
Justification of Cost: To undertake a high quality codesign study expert co-facilitation has been sourced from Bridges Self Management. This includes a member of the Bridges team to attend and co-facilitate 1 meeting at each of the 3 sites, to upskill Helena to develop the skills, knowledge and confidence required to facilitate and effectively deliver co-design events This includes preparation and debrief time, as well as travel expenses if required. Additionally includes for a member of the Bridges team to attend and co-facilitate the project launch event

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
	0	3200	0	0	3200

Description: Room hire and refreshments for cosdesign large hybrid group x 2
Type of cost: NHS
Please describe fully: Room hire and refreshments for cosdesign large hybrid group x 2
Justification of Cost: Room hire is free for a large meeting room at St George's Hospital with disabled access and audio-visual facilities. To include refreshments an estimation of £45 per meeting has been obtained.

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
	0	90	0	0	90

Description: Sumari JBI software
Type of cost: NHS
Please describe fully: Sumari JBI software
Justification of Cost: Specialist web-based software for scoping review based on JBI systematic review methodologies. I will be undertaking JBI scoping review training which will include software use. Current price is $130 USD/year (https://sumari.jbi.global/pricing), which is approximately £100 at current rates

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
	100	0	0	0	100

Description: Otter.ai transcription
Type of cost: NHS
Please describe fully: Otter.ai transcription
Justification of Cost: Otter Pro (https://otter.ai/) was recommended by Bridges Self Management. Otter.ai provides transcription services for the codesign interviews and groups which will save many hours of work. The annual subscription includes up to 90 min per conversation, 1200minutes (20 hours) monthly transcription and can link directly with Zoom. The subscription will also be utilised for PPI group meetings to ensure accurate records are kept and easily accessible.

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
	0	78	0	0	78

Description: codesign participant expenses
Type of cost: NHS
Please describe fully: codesign participant expenses
Justification of Cost: For the codesign methodology (Study 2), participation by nurses and people with CKD is essential. This will be done by means of small groups and additional individual interviews for those unable to participate in group sessions. Virtual participation is offered where possible to limit travel and time requirements, However, for the 2 large codesign meetings (Stage 2 and 4) travel costs for those who can attend in person may be incurred, (although the London daily travel cap applies). There is a possible need for travel via taxi for non-staff if they have physical accessibility requirements. Codesign stages and expense estimations (Year 2) Aiming to recruit 20-24 participants so the estimated budget is for 24 but may be reduced if 20 only recruited. Stage 1 (small groups/interviews 24 participants x £25) = £600 Stage 2 (large group 24 participants x £25) = £600 Stage 3 online 2 codesign groups of 8 persons x 2 meetings = 32 total contacts (32 x £25) = £800 Stage 4 final feedback/launch meeting 24 persons £600

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
	0	2600	0	0	2600

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
Other Direct Costs Total	100	7768	0	0	7868

Training And Development

Type of cost: NHS
Description: Short Courses
Item Description: Cardiff University JBI Scoping Review one day workshop (online)
Justification of Cost: This workshop will provide and introduction to the theories and concepts relating to scoping reviews and other types of evidence synthesis. It will give me the skills and knowledge to plan, undertake and report my scoping review following the internationally recognised Jill and equip participants with the knowledge and tools required to successfully plan for and undertake and report a scoping review following the Joanna Biggs Institute approach

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
	240	0	0	0	240

Type of cost: NHS
Description: Short Courses
Item Description: An Introduction to Qualitative Research Methods
Justification of Cost: Delivered by the Qualitative Health Research Centre (QUAHRC) at King's College London. Provides an overview and introduction to qualitative research methods, including individual interviews, focus group discussions, observation, ethical issues, and thematic analysis, NVivo and rigour in qualitative research. This will help prepare the co-design phase of the study and ensure I have a solid background in qualitative research to develop my skills especially within the context of the new regional renal research institute and other medical research environments. This is a reduced cost for Alumni and NHS Staff.

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
	750	0	0	0	750

Type of cost: NHS
Description: Short Courses
Item Description: Point of Care Foundation: Co-design Workshop
Justification of Cost: This will provide me with the foundation of codesign, prepare me for the specialist Bridges training and is essential for the co-design study and future research aspirations to develop codesign research within a medical research environment.

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
	0	395	0	0	395

Type of cost: NHS
Description: Short Courses
Item Description: Bridges Self-Management Narrative Interview Training
Justification of Cost: Bespoke narrative interview training (Small group - max 6 people) 2 x 90-minute workshops, highly contextualised training including facilitated learning, practise opportunities and examples of good practice, co-delivered by people with lived experience and clinical associate. Will support the co-design phase and enhance my research and clinical skills.

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
	0	400	0	0	400

Type of cost: NHS
Description: Short Courses
Item Description: Bridges Self Management Facilitation Training
Justification of Cost: Bespoke training (Max 6 people) 2 x 90-minute workshops, Interactive workshop style training including facilitated learning, practise opportunities and examples of good practice. Will build knowledge, skills and confidence in facilitation skills for both online and in-person sessions Co-delivered by people with lived experience and clinical associate. Will support the co-design phase and enhance my research, teaching and clinical skills.

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
	0	400	0	0	400

Type of cost: NHS
Description: Short Courses
Item Description: Bridges Supported Self-Management Training
Justification of Cost: Open course 8 hours of training fully contextualised to people living with long term conditions Co-delivered by people with lived experience and clinical associate 3 copies of Bridges Self-Management resource for a range of conditions. Will support the co-design phase and enhance my research and clinical expertise.

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
	0	600	0	0	600

Type of cost: NHS
Description: Short Courses
Item Description: Introduction to Implementation Science: 6 week online course
Justification of Cost: Discounted rates for St George's students ARC South London https://arc-sl.nihr.ac.uk/introduction-implementation-science-new-online-module The course covers the core scientific principles and investigative methodologies of implementation science, including quantitative, qualitative and econometric approaches. It provides theoretical and practical training and resources for the planning, development, implementation and evaluation of health and care interventions. It will consolidate my previous learning and prepare me to lead on the roll out of nutrition screening using both our inpatient and outpatient screening tools

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
	0	0	950	0	950

Type of cost: NHS
Description: Short Courses
Item Description: Implementation Science Masterclass
Justification of Cost: Discounted rates for St George's students https://www.kcl.ac.uk/short-courses/implementation-science-masterclass-1 Complementary to the Introduction in Implementation Science course, this is a two-day course to advance my understanding and skills in implementation science. This will provide me with a solid foundation in implementation science to support the current research proposal and future implementation of nutrition screening practices nationally and internationally.

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
	0	0	400	0	400

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
Training And Development Total	990	1795	1350	0	4135

Total Direct Costs

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
Total Direct Costs	64265	76452	71728	74773	287218

Indirect costs

HEI Indirect costs
No costs have been added

Commercial Indirect Costs
No costs have been added

Other Partner Organisation Indirect costs
No costs have been added

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
Total Indirect Costs	0	0	0	0	0

Research Costs Requested From Funder

	Direct costs	Indirect costs/Overheads	Amount requested

Total Higher Education Institution Costs (80%)	0	0	0
Total Higher Education Institution Costs (100%)s	0	0	0
Total Commercial Costs	0	0	0
Total Other Parther Organisation Costs	0	0	0
Total NHS Research Costs	287218	0	287218
Total Research Costs Requested From Funder	287218	0	287218

Summary

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
Salaries	**61487**	**64561**	**67789**	**71178**	**265015**
Travel-subsistence-and-conference-fees	**0**	**100**	**100**	**2765**	**2965**
Equipment	**628**	**300**	**400**	**0**	**1328**
Consumables	**200**	**390**	**390**	**170**	**1150**
Patient-and-Public-Involvement	**860**	**1538**	**1699**	**660**	**4757**
Other-Direct-Costs	**100**	**7768**	**0**	**0**	**7868**
Training And Development	**990**	**1795**	**1350**	**0**	**4135**
HEI-Indirect-costs	**0**	**0**	**0**	**0**	**0**
Commercial-Indirect-Costs	**0**	**0**	**0**	**0**	**0**
Other-Partner-Organisation-Indirect-costs	**0**	**0**	**0**	**0**	**0**
Total	**64265**	**76452**	**71728**	**74773**	**287218**

NHS Support Costs

Cost Item:	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
Codesign and validation study support costs	0	1224	7089	0	8313.0

Justification:
Codesign and validation study support costs: The SoECAT shows costs for research support for consenting and data collection for studies 2 (year 2) and 3 (year 3) 1 and 2. Additional research costs have been calculated on the SoECAT but will not be incurred as events will be performed by the central research team (ie. the lead applicant) so are not considered as a research support cost.

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
NHS Support Costs Total	0	1224	7089	0	8313

Have you discussed and agreed these costs with the Lead Network?	Yes

NHS Treatment Costs
No costs have been added

Have you discussed and agreed the NHS costs with the Lead Trust?	Yes

Is the patient/service user care being provided different from the usual standard treatment for the condition?
Yes

Usual Treatment Costs
No costs have been added

Excess Treatment costs/(savings)

	Cost per patient(£)	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
Total NHS Excess Treatment costs/(savings)	0	0	0	0	0	0

	Total (£)
NHS Support Costs requested from networks	8313
NHS Excess Treatment costs/savings	0
TOTAL NHS SUPPORT AND TREATMENT COSTS	8313

Total Funding Required (including non-research costs)

	Total (£)
Total Research Costs Requested from Funder	287218
Total NHS (Support and Excess Treatment) Costs	8313
Total Funding Required	295531

10. Uploads

Please upload the following documents, if necessary.

When uploading a document to your application, please ensure that it is in an appropriate file format. It is not possible to upload JPEG images, and with the exception of the SoECAT Form, which should be uploaded as an Excel spreadsheet, all files should be uploaded in Adobe PDF or Word DOC/DOCX format. Please refer to the Guidance Notes for further information on the attachments required.

(References - NIHR DCAF References A4.docx) is included as an appendix within this file.

(Figures / Tables - Upload CKD oNUT single A4.pdf) is included as an appendix within this file.

(Research Timetable - DCAF Research Timetable HJ.pdf) is included as an appendix within this file.

(SoECAT Template - Funder Export_344985_SoECAT_CKD oNUT2.xlsx) is included as an appendix within this file.

11. Participants and Signatories

Participants
You are required to supply the name and email address (if not already registered on the ARAMIS application system) of each required participant. Everyone included in this section will be acting as a 'participant' in your application and must agree to be named as such.

By confirming participation, participants are acknowledging their involvement and input into this application, and are agreeing to be involved, as indicated, in any resultant fellowship.

Once the 'Administrative Authority or Finance Officer' participant has confirmed their participation, they will be able to access the 'Detailed Budget' section of the application form. You should collaborate with your 'Administrative Authority or Finance Officer participant to ensure that this section is completed satisfactorily.

You must ensure that all participants are happy for your application to be submitted before you submit it.

It is permissible for a single individual to serve as both a signatory and a participant, but they will need to be added separately under each heading.

Named participants must confirm their participation on your application before you will be able to select the 'SUBMIT' option. Please refer to the Submission Process Flow Diagram and the Applicant Guidance Notes for further information.

Primary Doctoral Supervisor

Doctoral Primary Supervisor Participant 1

Title	Professor
Forename(s)	Fiona
Surname	Jones
Position	Professor Rehabilitation Research

Additional Academic Supervisor(s)

Supervisor Participant 1

Title	Dr
Forename(s)	Sue
Surname	Green
Position	Nursing and Clinical Sciences

Supervisor Participant 2

Title	Dr
Forename(s)	Helen
Surname	MacLaughlin
Position	Associate Professor

Practice Supervisor(s)

Practice Support Participant 1

Title	Professor
Forename(s)	Debasish
Surname	Banerjee
Position	Clinical Lead

Practice Support Participant 2

Title	Ms
Forename(s)	MISS GEMMA
Surname	STOTT
Position	

Administrative Authority or Finance Officer

Administrative Authority or FO Participant

Title	Mr
Forename(s)	Oliver
Surname	Palmer
Position	Research Funding Officer

Signatories

You are required to supply the name and email address (if not already registered on the ARAMIS application system) of each required 'Head of Department / Senior Manager' signatory. Once their contact details have been entered below, these signatories can be invited to log in and confirm their participation in your application.

Once the nominated signatories have confirmed their participation, they can access and complete the required 'Host Organisation Statement of Support' in the 'Training & Development and Research Support' section of the application form.

Your signatories must confirm their participation in your application and complete the section they are responsible for before you can select the 'submit' option.

Signatories must also approve your application after you select the 'submit' option but BEFORE the application submission deadline.

Signatory approval from your 'Head of Department / Senior Manager' signatories will result in the application being fully submitted to the NIHR. All parties (applicant, participants and signatories) will be notified of this via an automated system-generated email.

The NIHR will not accept your application unless fully approved by your signatories before the 1 p.m. deadline on Tuesday 11th June 2024.

Heads of Departments / Senior Managers

Head of Department or Senior Manager 1

Title	Professor
Forename(s)	Charlotte
Surname	Clark
Position	Institue Director

Head of Department or Senior Manager 2

Title	Professor
Forename(s)	Heather
Surname	Jarman
Position	Research Director for Nurses, Midwives and AHPs / Emergency Department Academic Lead

12. Acknowledgement, Review and Submit

Conflict checks: Please declare any conflicts or potential conflicts of interest that you may have in undertaking this research, including any relevant, non-personal & commercial interest that could be perceived as a conflict of interest.
None

By completing this form, the programme reserves the right to share, in confidence, details of your application with other parts of the NIHR and with the Department of Health and Social Care.

Agreement to Terms and Conditions: I confirm that I have secured all necessary licences and ethical approvals in relation to the research and will abide by the terms of those licences and approvals in carrying out the research. I confirm that I have read the NIHR Carbon Reduction Guidelines and, where possible, have taken steps to reduce the carbon emissions generated by this research. I confirm that all material within this application, with the exception of sections that other participants are required to complete, is my own work and I have taken steps to ensure that no parts of the application have been inappropriately plagiarised. I undertake to conduct the research according to such guidelines for good research practice as the Host Organisation may from time to time lay down according to the NIHR research governance framework. I undertake to conduct the research according to such guidelines for good research practice as the Host Organisation may from time to time laydown according to the UK Policy Framework for Health and Social Care Research. In completing this, I declare that, to the best of my knowledge, the information provided in this application, is true, accurate and complete. I have read the Guidance Notes and agree to accept the process by which an application is assessed and agree to abide by the conditions under which an award may be granted.
☒Confirmed

Appendices

1. NIHR DCAF References A4.docx
2. Upload CKD oNUT single A4.pdf
3. DCAF Research Timetable HJ.pdf
4. Funder Export_344985_SoECAT_CKD oNUT2.xlsx

Appendix #1: NIHR DCAF References A4.docx

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Appendix #2: Upload CKD oNUT single A4.pdf

Figure 1. Causes and clinical implications of protein-energy wasting in CKD (Adapted from: Chan 2021)

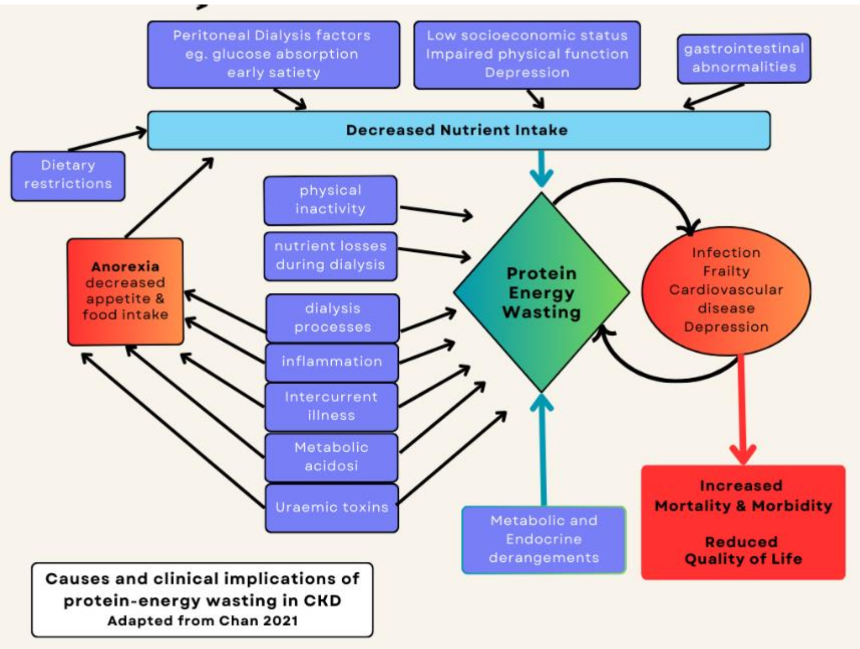

Figure 2. Renal iNUT Screening Tool (Jackson et al 2019)

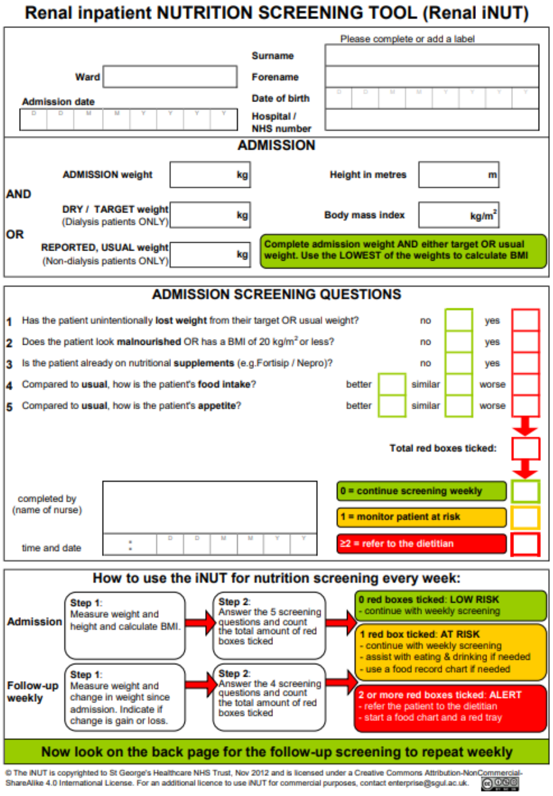

FIGURE 3. Logic Model

Target population	Inputs	Activities	Key Outputs	Initial Outcomes	Impacts
Adults with CKD stages 4-5 attending renal outpatient services Specialist renal nurses working in renal outpatient services	• Experienced renal dietitian time (HJ) • Researcher time and training (HJ) • Supervisor/collaborator time/expertise • Additional student input (MSc/ professional course) under supervision of HJ • Co-design nurse and PwCKD participant time and contributions • Codesign participant reimbursement • PPI Consultation and reimbursement • Onsite launch and MDT training sessions • Validation study participant time • Trained renal nurses completing screening • Promotion and information for PwCKD and MDT • Relevant IT software	• Ethics clearance • Conduct scoping review • Clinical trial registration • Recruit participants • Codesign meetings • Transcription, summary, analysis of codesign meetings • Implement and monitor NST in selected outpatient services in 3 hospitals • Nutrition screening • Dietetic assessments • Collection of demographic and descriptive data • Duplicate screening • Acceptability and satisfaction assessments • Dissemination of findings	**Co-design** • Validated Nurse NST for kidney outpatient settings, • Implementation process • Nurse training resource • First line advice for nurses and PwCKD (for a given NST result) • PwCKD self-management resource **Dissemination** • PwCKD local and national organisation publications • Social media output • MDT publications • Peer-reviewed journal • Conference presentations • Dedicated website and registration for clinicians and researchers wishing to use NST /co-designed output	• Nurses receiving nutrition screening training • Co-designed resources distributed • % patients receiving nutrition screening • % patients with nutrition care plan • Dietitian referral rates • Reported prevalence of malnutrition in outpatient services and relationship to frailty, complications and other monitored parameters	**Advancing knowledge and research** • No. direct publications presentations, and citation statistics • Usage of oNUT in studies of malnutrition and nutrition intervention in CKD • No. cultural/translated adaption of oNUT and codesign resources **Capacity/capability building** • Improved nurse training and skills including referrals to dietitians • Increased patient use of prevention /self-management strategies • Efficient distribution of dietitian workforce **Practice impact** • Adoption or refining of oNUT and resources beyond trial (CKD and/or other specialist services) **Policy impact** • Referenced in national/international guidelines **Community /Patient impact** • Better understanding of malnutrition and self-management opportunities • Improved quality of life • Reduced frailty • Reduced hospitalisation • Improved access to dietitian for malnutrition

Culture, native/home language	Socioeconomic status, education level (PwCKD)	outpatients staffing and environment	CKD stage/treatment, co-morbidities and current health status (PwCKD)	Service support (clinical, managerial, senior leadership)
Personal experience of malnutrition (self or family)	Motivation/engagement	Perceived value and relevance	Experience/seniority (nurses)	Satisfaction and acceptability

CKD: chronic kidney disease, NST: Nutrition Screening Tool, MDT: multidisciplinary team, oNUT: codesigned outpatient nutrition screening tool, PPI: patient and public involvement, PwCKD: people living with CKD

Appendix #3: DCAF Research Timetable HJ.pdf

YEAR			Year 1 (April 25-March-26)				Year 2 (April-25-March-26				Year 3 (April26-March 27				Year 4 (April 27-		
Quarter		Q -1	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3
Set up																	
Study 1. Scoping Review	PPI group input																
	Literature searching, selection, analysis and review and conclusions																
	Write and submit for publication																
	Plain English summary for codesign stage with input from PPI group																
Study 2. Codesign	Ethics and regulatory approvals																
	Develop PPI sheets and consent forms with PPI and research team																
	Determine interview/group structure, topic guide and questions with PPI and research team																
	Recruitment by purposeful sampling																
	Codesign groups and interviews																
	Data analysis																
	Write up and submit for publication with input from PPI and research team																
Study 3. Feasibility	Ethics and regulatory approvals Purposeful sampling Recruitment Data collection; semi-structured interviews Data analysis Write up and submit for publication with input from PPI group and research team Summarise for co-design workshop with input from PPI group																
	Develop PPI sheets and consent forms with PPI group and research team																
	Introduce screening tool to clinics with nurses training																
	Ongoing dialogue and support for nurses and relevant MDT																
	Recruitment by purposeful sampling																
	Validation assessments																
	Data analysis																
	Write up and submit for publication with input from PPI group and research team																
Scientific Conferences (UKKW, ESPEN, ANZSA																	
Research visit																	
Training	Graduate school skills program																
	JBI Scoping Review 1 day workshop																
	Public Involvement in Research																
	KCL An Introduction to Qualitative Research Methods July																
	Point of Care Foundation: Co-design Workshop (currently schedules sept-Nov only)																
	Bridges Training & faciliation																
	NIHR ARC Research Leadership Academy Jan-June (1 hour per week)																
	Implementation Science course and masterclass																
	The power of infographics in research dissemination																
Clinical practice																	
Deliverables	Publications																
	Annual project reports																
	Conference presentations																
	PhD thesis write up																

Appendix #4: Funder Export_344985_SoECAT_CKD oNUT2.xlsx